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DREAM LENS

Sleep Tracking

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ABSTRACT

Problem: Manual sleep stage classification from Electroencephalogram (EEG) signals remains a labour-intensive, subjective, and expensive process that creates a systemic bottleneck in the diagnosis of sleep disorders. Existing automated deep learning approaches are collectively trapped in a persistent accuracy-versus-efficiency dichotomy: Recurrent Neural Network (RNN) benchmarks such as DeepSleepNet achieve practical efficiency but suffer from the vanishing gradient problem, limiting their ability to model long-range temporal dependencies across a full night of sleep, whilst standard Transformer architectures resolve this contextual limitation at a computational cost that precludes deployment on wearable, resource-constrained hardware. This project addresses the critical gap for a Lightweight Transformer architecture that delivers Transformer-level contextual modelling at RNN-level computational cost, integrated within a fully deployed, consumer-accessible sleep intelligence application.

Methodology: This research follows a quantitative, deductive, and experimental methodology. The Sleep-EDF database (Kemp et al., 2000), hosted on PhysioNet, was utilised as the universal benchmark dataset. A robust preprocessing pipeline was constructed using MNE-Python, applying a 0.3–35 Hz FIR bandpass filter, resampling to 100 Hz, segmenting into 30-second non-overlapping epochs, and merging the legacy R&K N4 stage into N3 per AASM standards. Phase 1 validated a CNN-LSTM hybrid baseline (DreamLens Alpha). Phase 2 delivered the primary contribution: a Lightweight Transformer encoder (2 layers, 4 attention heads, $d_{\text{model}}=128$, ~1.8M parameters) trained with Weighted Cross-Entropy Loss and evaluated under subject-wise k-fold cross-validation. The complete DreamLens system was deployed as a full-stack web application comprising an asynchronous FastAPI backend, MongoDB persistence layer, Next.js 14 consumer dashboard with Recharts interactive charts, Clerk authentication, and two distinct consumer contributions: a standalone AI Sleep Coach powered by the Groq API (llama-3.1-8b-instant) and one-click jsPDF clinical sleep reports.

Targets and Contribution: The DreamLens Lightweight Transformer is designed to achieve Overall Accuracy exceeding 85% and Macro F1-Score exceeding 80% on the Sleep-EDF dataset under subject-wise cross-validation surpassing the CNN-LSTM baseline (ACC 81.4%, MF1 74.1%) and the published DeepSleepNet benchmark (ACC ~82%, MF1 ~76.9%). The system contributes approximately 1.8 million parameters and approximately 4 milliseconds

per-epoch CPU inference time, satisfying wearable-grade deployment feasibility. The DreamLens full-stack platform integrating ML inference, derived clinical metrics (Sleep Score, WASO, REM Latency, Sleep Cycles), multi-chart visualisation, and context-grounded generative AI coaching constitutes a novel application-layer contribution demonstrating a concrete pathway from academic sleep staging research to consumer health intelligence.

Subject Descriptors:

Computing methodologies → Machine learning → Neural networks → Transformer architectures

Applied computing → Life and medical sciences → Health informatics

Human-centred computing → Web-based interaction

Keywords: Sleep Stage Classification, EEG, Lightweight Transformer, Deep Learning, Sleep Health Informatics, Generative AI, Full-Stack Consumer Application

CHAPTER 01: INTRODUCTION

1.1 Chapter overview

This opening chapter establishes the complete foundational context for the DreamLens project an investigation at the convergence of deep learning, biomedical signal processing, and consumer health technology. The chapter begins by situating the research within the clinical landscape of sleep medicine, articulating both the limitations of the gold-standard manual Polysomnography (PSG) scoring process and the inadequacies of existing automated computational approaches. The central technical problem the accuracy-versus-efficiency trade-off is then precisely defined, and the motivation for addressing it is articulated from both a healthcare equity and a technical advancement perspective.

The chapter documents the project's two-phase evolution: Phase 1 designed and validated a CNN-LSTM hybrid baseline model (DreamLens Alpha) to establish the performance floor and validate the data engineering pipeline; Phase 2 delivered the primary research contribution the DreamLens Lightweight Transformer integrated within a production-grade full-stack consumer application. The research gap, contribution to the body of knowledge, three research questions, research aim, and ten research objectives (RO1–RO10) are all formally defined and mapped. Crucially, the AI Sleep Coach is recognised as a standalone third contribution (RQ3, RO9), distinct from the core ML model (RQ1) and the system deployment (RQ2). The chapter concludes by delineating the project scope and providing the hardware and software requirements that governed the development environment.

1.2 Problem background

Sleep is a fundamental biological process underpinning cognitive performance, emotional regulation, memory consolidation, immune function, and long-term cardiovascular health. The World Health Organisation has classified insufficient sleep as a global public health epidemic, with sleep disorders including insomnia, obstructive sleep apnoea, restless leg syndrome, and narcolepsy affecting an estimated one-third of the adult population in developed nations. Left undiagnosed and untreated, these conditions are strongly associated with elevated risks of hypertension, type 2 diabetes, clinical depression, and neurodegeneration, including early-onset Alzheimer's disease. The direct and indirect economic burden of sleep disorders through

lost workplace productivity, healthcare utilisation, and road traffic accidents attributable to sleep-deprived driving runs to hundreds of billions of dollars annually in major economies.

The established clinical standard for diagnosing sleep disorders is Polysomnography (PSG) a comprehensive overnight physiological recording conducted in specialist sleep laboratories. A full PSG session captures multiple biological signals simultaneously: the Electroencephalogram (EEG), which records electrical brain activity via scalp electrodes and is the primary signal for sleep staging; the Electrooculogram (EOG), which tracks eye movements; the Electromyogram (EMG), which measures muscle tone; and additional cardiorespiratory channels. Of these, the EEG is considered paramount, as the distinct brainwave oscillations it encodes Delta waves (0.5–4 Hz) during deep slow-wave sleep, Sleep Spindles (12–14 Hz) and K-complexes during N2, Theta waves (4–8 Hz) during N1 and REM, and Alpha waves (8–13 Hz) during relaxed wakefulness provide the physiological basis for classifying sleep stages.

In clinical practice, trained sleep technicians manually review each full-night EEG recording, segmenting the continuous signal into contiguous 30-second windows known as epochs and assigning each to one of five stages per the American Academy of Sleep Medicine (AASM) guidelines: Wake (W), Non-REM Stage 1 (N1), Non-REM Stage 2 (N2), Non-REM Stage 3 (N3 deep or slow-wave sleep), and Rapid Eye Movement (REM) sleep. A typical 8-hour recording produces over 900 individual epochs, and the complete manual scoring process demands between two and four specialist hours per patient. This bottleneck is simultaneously labour-intensive, expensive (USD 1,500–3,000 per in-clinic PSG session), geographically concentrated (specialist sleep laboratories cluster in urban centres), and subject to documented inter-rater variability in which different trained technicians may score the same ambiguous epoch differently.

The mounting demand for scalable, accurate, and affordable sleep analysis has driven two decades of research into automated deep learning systems. Early CNN-based epoch-wise models automated feature extraction but remained memoryless treating each epoch in isolation without encoding the sequential context of sleep architecture. The introduction of RNNs particularly LSTM networks via the influential DeepSleepNet model incorporated temporal memory but with structural limitations that directly motivate this project's core research contribution, as elaborated in Section 1.3.

1.3 Problem definition

The core technical challenge in automated sleep staging is the accurate modelling of temporal dependencies—the structured, biologically governed relationships between the current sleep epoch and the broader context of preceding and subsequent epochs throughout the night. Sleep is not a random sequence of physiological states; it follows a regular cyclical architecture of approximately 90-minute cycles in which the brain progresses through NREM and REM stages in a characteristic pattern. This architecture encodes clinically critical information: REM latency (time from sleep onset to first REM), deep sleep proportion, and sleep cycle count are all standard diagnostic markers whose computation requires system-level understanding of the full night's progression, not epoch-by-epoch classification.

Recurrent Neural Networks—specifically LSTM networks—were designed to address temporal dependency modelling by maintaining a hidden state vector that encodes prior context. The landmark DeepSleepNet model (Supratak et al., 2017) pioneered the hybrid CNN-LSTM approach, achieving an overall accuracy of approximately 82% and a Macro F1-Score of 76.9% on Sleep-EDF. However, RNNs carry a well-documented structural limitation: the vanishing gradient problem causes the learning signal to degrade as it propagates backwards through time, restricting effective sequential memory to approximately 20–30 epochs (10–15 minutes of sleep). This means LSTM-based models cannot reliably connect the sleep state at 11 pm to its clinically significant context at 3 am—precisely the temporal scale at which the distinction between first-cycle and late-cycle REM carries diagnostic weight.

Transformer architectures, introduced by Vaswani et al. (2017) and subsequently revolutionising Natural Language Processing, address this limitation through the Self-Attention mechanism. By computing weighted relevance scores between every pair of positions in a sequence simultaneously—rather than processing recurrently—the Transformer captures global context without the information bottleneck of a fixed-size hidden state. SleepTransformer (Phan et al., 2022) confirmed that this translates to measurable performance gains over RNN benchmarks on Sleep-EDF (ACC ~88.5%, MF1 ~83.2%). However, standard Transformer architectures introduce their own limitation: computational cost scales quadratically with sequence length, and large parameter counts make full Transformer models prohibitively expensive for the battery-constrained, resource-limited hardware of consumer wearable devices. This creates the defining dichotomy: efficient but contextually limited RNNs versus contextually powerful but computationally heavy Transformers. There is a critical, insufficiently addressed need for a unified solution that achieves Transformer-level context at wearable-grade computational cost.

A second gap entirely unaddressed by the existing academic literature is the absence of any end-to-end consumer application that translates the raw epoch-level classifications of a lightweight EEG model into clinically meaningful derived metrics, interactive visualisations, and personalised generative AI coaching. DreamLens was designed to fill both gaps simultaneously.

1.3.1 Problem statement

Current automated sleep staging models face an unresolved dichotomy: Recurrent Neural Network architectures are computationally efficient but structurally incapable of modelling long-range temporal dependencies in EEG signals due to the vanishing gradient problem, whilst standard Transformer architectures overcome this contextual limitation at a computational cost that renders them impractical for deployment on resource-constrained wearable devices. Furthermore, no existing academic work has integrated a lightweight EEG classification model within a full-stack consumer application that transforms raw predictions into clinically meaningful, AI-augmented health intelligence accessible to non-expert users.

1.4 Research motivation

The primary motivation for DreamLens is the urgent and growing need to democratise access to meaningful sleep health analysis. Sleep disorders are a global epidemic, yet the diagnostic pathway centred on expensive, clinic-based, manually scored PSG creates a systemic access barrier that disproportionately affects lower-income populations and under-resourced healthcare systems. By engineering a lightweight automated system capable of running on consumer hardware and delivering insights through an accessible web interface, this research is directly motivated by the potential to shift sleep health from a reactive, clinic-dependent model to a proactive, home-based monitoring paradigm.

From a technical perspective, the project is motivated by the active research frontier at the convergence of efficient deep learning and biomedical signal processing. The field is in a transition period: the first generation of Transformer-based sleep staging models has proven the capability of self-attention for EEG but exposed the efficiency gap, and a new wave of lightweight Transformer research including SleepSatelightFTC (Ito & Tanaka, 2025) is emerging to resolve it. DreamLens positions itself directly within this cutting-edge trajectory. The AI Sleep Coach contribution is independently motivated by the demonstrated consumer demand for AI-driven health personalisation confirmed by the 66.1% positive response to

coaching features in the DreamLens user survey and by the novel opportunity to apply a context-grounded LLM to sleep health coaching, where no comparable work exists in the literature.

1.5 Existing work

The evolution of automated sleep staging reveals five principal architectural approaches, each representing a distinct position on the accuracy-versus-efficiency continuum. Table 1-1 provides a structured summary of these key works, which are reviewed in full critical detail in Chapter 2.

Citation	Architecture Summary	Limitation	Contribution to DreamLens
Supratak et al. (2017) DeepSleepNet	CNN-LSTM hybrid: CNN extracts within-epoch features; LSTM models sequential transitions between epochs on Sleep-EDF.	LSTM vanishing gradient limits effective memory to approximately 20–30 epochs; fails on long-range sleep-cycle context.	Primary performance baseline (ACC 82%, MF1 76.9%) that the DreamLens CNN-LSTM Phase 1 model replicates and the Lightweight Transformer must surpass.
Phan et al. (2022) SleepTransformer	Full Transformer encoder applied to sequences of EEG epochs; self-attention captures global context across the full night.	Large parameter count (~4.5M) and quadratic attention complexity render the model computationally heavy and impractical for wearable deployment.	Validates global self-attention for sleep EEG; sets the accuracy ceiling (ACC 88.5%, MF1 83.2%) DreamLens aims to approach with far fewer parameters.
Jia et al. (2023) Efficient 1D-CNN	Optimised 1D convolutional architecture achieving competitive accuracy without recurrent or attention layers on Sleep-EDF.	Epoch-wise classification no mechanism for global temporal context modelling across the full sleep recording.	Establishes the efficiency benchmark: DreamLens must match this model's computational cost whilst adding global attention capability.
Ito & Tanaka (2025) SleepSatelightFTC	Lightweight, interpretable Transformer for single-channel EEG staging with reduced head count and layers vs. SleepTransformer.	Limited benchmarking against CNN-LSTM hybrids; not deployed within any consumer-facing application.	Primary architectural inspiration for the DreamLens Lightweight Transformer; validates the feasibility of the lightweight Transformer approach.
GNN Approaches (Zeng et al. and others)	Graph Neural Networks model the brain as a spatial graph, learning	Requires multi-channel EEG input as an architectural prerequisite;	Confirms that the single-channel constraint is a deliberate and well-justified

Citation	Architecture Summary	Limitation	Contribution to DreamLens
	spatio-temporal relationships between EEG electrodes.	incompatible with the single-channel wearable constraint.	scope limitation; multi-channel GNN approaches are explicitly out of scope.

Table 1-1: Existing Work Summary Automated Sleep Staging Architectures

A critical reading of these five approaches reveals that none has simultaneously addressed both the efficiency requirement for wearable deployment and the accessibility requirement for non-expert consumer use. DeepSleepNet remains the dominant benchmark but its LSTM memory fails at full sleep-cycle scale. SleepTransformer achieves superior accuracy at prohibitive parameter cost. The Efficient 1D-CNN optimises for speed at the cost of global context. SleepSatelightFTC is the closest architectural predecessor but exists only as an academic model without any consumer deployment. DreamLens addresses the collective gap by coupling a Lightweight Transformer optimised to retain self-attention at minimal parameter cost with a fully deployed, AI-augmented consumer application whose Sleep Coach is itself a novel contribution in the health AI domain.

1.6 Research gap

Two distinct and insufficiently addressed gaps define the specific focus of this research. The first is the Context-versus-Efficiency Gap. The sleep staging literature is polarised between RNN models that are wearable-deployable but epistemically limited, and Transformer models that are contextually powerful but computationally prohibitive. Lightweight Transformer variants have begun to emerge (Ito & Tanaka, 2025; Li, 2024; Kim et al., 2023) but the body of rigorous comparative benchmarking against CNN-LSTM hybrids on the single-channel Fpz-Cz Sleep-EDF configuration remains sparse. DreamLens addresses this gap through a controlled, head-to-head comparison under identical experimental conditions.

The second is the Model-to-Application Gap. The overwhelming majority of academic research in sleep staging produces model weights, evaluation tables, and paper submissions but no deployed, end-to-end consumer application that translates the model's epoch-level predictions into a form that is accessible, interpretable, and actionable for a non-expert user. No existing work has integrated a lightweight EEG classifier with a full-stack web application featuring persistent authenticated user history, interactive multi-chart visualisation, and a

context-aware generative AI coach whose responses are grounded in the user's specific sleep session data. DreamLens fills this gap entirely.

1.7 Contribution to body of knowledge

This research makes three distinct and complementary contributions. The primary research contribution is the design, training, and rigorous benchmarking of the DreamLens Lightweight Transformer – a novel architecture for single-channel EEG sleep staging targeting Overall Accuracy exceeding 85% and Macro F1-Score exceeding 80% on Sleep-EDF under subject-wise cross-validation, with approximately 1.8 million parameters and approximately 4 milliseconds per-epoch CPU inference time satisfying wearable deployment constraints. The controlled comparison against the CNN-LSTM baseline under identical experimental conditions provides clean, unconfounded empirical evidence for the performance advantage of self-attention over LSTM sequence modelling.

The second contribution is the DreamLens full-stack consumer sleep intelligence platform – the first in the sleep staging literature to demonstrate end-to-end integration of a lightweight EEG classifier with a production-grade consumer application. The platform introduces a novel Sleep Score computation algorithm (a clinically grounded weighted composite of sleep efficiency, deep sleep percentage, REM percentage, WASO penalty, and cycle completeness), three interactive Recharts visualisations, and persistent authenticated session history, demonstrating a concrete and reproducible pathway from academic model to consumer health tool.

The third contribution – treated as a standalone contribution – is the DreamLens AI Sleep Coach: the first context-grounded generative AI coaching system for sleep health, where the user's specific sleep session JSON (Sleep Score, WASO, REM Latency, Stage Distribution) is injected directly into the LLM context window of the Groq llama-3.1-8b-instant model, producing personalised, data-grounded coaching responses distinct from generic AI health chatbots. This context injection architecture has broad applicability across consumer health monitoring domains beyond sleep staging.

1.8 Research challenges

The development of DreamLens encountered significant challenges across the data engineering, machine learning, and software engineering dimensions. The most persistent data-level challenge was the severe class imbalance in the Sleep-EDF dataset: the N2 stage constitutes approximately 47% of all epochs whilst N1 represents only 5–8%. A naive training

approach produces deceptively high overall accuracy through systematic N2 dominance whilst failing on the clinically important minority stages (N1, N3, REM). This was addressed through Weighted Cross-Entropy Loss with inverse-frequency class weights, and through the adoption of Macro F1-Score as the primary evaluation metric.

At the model architecture level, the CNN-LSTM baseline empirically confirmed the core research hypothesis during Phase 1: the LSTM exhibited systematic misclassification at transition boundaries – particularly N2-to-REM transitions – and demonstrated a characteristic pattern of N1-Wake confusion that is consistent with insufficient long-range context. These specific failure modes directly motivated and validated the architectural shift to the Lightweight Transformer. The design of the Transformer required careful parameter management to remain lightweight (approximately 1.8 million parameters) whilst retaining sufficient attention heads and layers for global context modelling. At the system integration level, asynchronous processing of large .edf files (often exceeding 50 MB) within FastAPI required careful implementation of thread pool execution to prevent server timeouts, and the design of the Groq context injection strategy required managing LLM context window limits by summarising session data into compact derived metrics rather than transmitting the full raw prediction array.

1.9 Research questions

The research is structured around three primary questions that map directly to the project's three principal contributions:

RQ1 (ML Architecture): Does a novel Lightweight Transformer architecture, trained on single-channel EEG (Fpz-Cz) from the Sleep-EDF dataset under subject-wise cross-validation, achieve Overall Accuracy exceeding 85% and Macro F1-Score exceeding 80%, surpassing both the CNN-LSTM baseline and the published DeepSleepNet benchmark, whilst maintaining a parameter count and per-epoch inference time compatible with wearable-grade deployment?

RQ2 (Full-Stack System): Is it feasible to transform raw epoch-level EEG classification outputs from a lightweight Transformer model into clinically meaningful consumer-accessible sleep intelligence – comprising derived health metrics, interactive data visualisations, authenticated session history, and downloadable clinical reports – within a production-grade full-stack web application?

RQ3 (AI Sleep Coach – Standalone Contribution): Can a context-grounded generative AI coaching system, powered by the Groq API (llama-3.1-8b-instant) with the user's specific sleep

session JSON injected into the LLM context window, deliver meaningfully personalised sleep coaching responses that are qualitatively more relevant and actionable than generic AI health chatbot interactions?

1.10 Research aim

The aim of this research is to design, develop, and evaluate DreamLens – an automated sleep intelligence system that resolves the accuracy-versus-efficiency trade-off in EEG-based sleep stage classification through a novel Lightweight Transformer architecture, that deploys those insights within a production-grade full-stack consumer web application with derived clinical metrics, interactive visualisations, and authenticated session history, and that delivers a third standalone contribution in the form of a context-grounded generative AI Sleep Coach powered by the Groq API.

1.11 Research objectives

The ten research objectives mapping the full delivery of this aim are presented in Table 1-2, with phase, research question mapping, and completion status.

Phase	ID	Research Objective	RQ	Status	Ch.
Phase 1 (Baseline)	RO1	Conduct a critical literature review of existing sleep staging architectures, establishing DeepSleepNet and Efficient 1D-CNN as primary performance benchmarks.	RQ1	Completed	Ch.2
	RO2	Implement a robust MNE-Python preprocessing pipeline for Sleep-EDF data, applying a 0.3–35 Hz FIR bandpass filter, resampling to 100 Hz, 30-second epoching, and AASM N4→N3 label standardisation.	RQ1	Completed	Ch.7
	RO3	Design and train a CNN-LSTM Hybrid Baseline Model (DreamLens Alpha) on Sleep-EDF, achieving a minimum classification accuracy of 80% to validate the pipeline and establish a performance floor.	RQ1	Completed	Ch.8
	RO4	Develop a functional end-user dashboard (Phase 1: Streamlit prototype) visualising the model's classification	RQ2	Completed	Ch.7

Phase	ID	Research Objective	RQ	Status	Ch.
		output as a Hypnogram and generating downloadable clinical PDF reports.			
Phase 2 (Final)	RO5	Design and train a Lightweight Transformer architecture (DreamLens Final), replacing the LSTM with a streamlined Multi-Head Self-Attention mechanism targeting ACC > 85% and MF1 > 80% on Sleep-EDF.	RQ1	Completed	Ch.6,7
	RO6	Quantitatively benchmark the Lightweight Transformer against the CNN-LSTM baseline and published benchmarks (DeepSleepNet, SleepTransformer, Efficient 1D-CNN) using ACC, MF1, Cohen's Kappa, and per-class F1-Scores under subject-wise k-fold cross-validation.	RQ1	Completed	Ch.8
	RO7	Develop an asynchronous FastAPI backend integrating PyTorch inference, derived clinical metric computation (Sleep Score, WASO, REM Latency, Sleep Cycles), and MongoDB session persistence.	RQ2	Completed	Ch.7
	RO8	Construct a production-grade consumer dashboard using Next.js 14 and Tailwind CSS with Clerk user authentication and three interactive Recharts visualisations: Hypnogram AreaChart, Stage Distribution PieChart, and Brainwave Power RadarChart.	RQ2	Completed	Ch.7
	RO9	Design and deploy a standalone AI Sleep Coach contribution powered by the Groq API (llama-3.1-8b-instant), injecting the user's specific sleep session JSON into the LLM context window and rendering personalised coaching responses via react-markdown in a floating chat widget.	RQ3	Completed	Ch.7
	RO10	Implement one-click downloadable clinical sleep reports using jsPDF, containing all derived metrics, health flag indicators, and a mandatory medical disclaimer.	RQ2	Completed	Ch.7

Table 1-2: Research Objectives (RO1–RO10) Phase, RQ Mapping, and Status

1.12 Project scope

1.12.1 In scope

The following define the boundaries of this research. Dataset: the Sleep-EDF public database (Kemp et al., 2000) both the SC cohort (40 recordings, 20 healthy subjects aged 25–34) and the ST cohort (78 recordings, 78 subjects aged 25–101) is the exclusive data source, ensuring direct benchmark comparability with all seventeen reviewed papers. Input signal: the analysis is strictly limited to single-channel EEG (Fpz-Cz), deliberately simulating the data availability of consumer wearable devices. Classification standard: all outputs conform to the five-stage AASM convention (W, N1, N2, N3, REM) with mandatory N4→N3 merge during preprocessing. Model architectures: Phase 1 CNN-LSTM hybrid for baseline validation; Phase 2 Lightweight Transformer for the primary research contribution. Full-stack system: FastAPI backend, MongoDB persistence, Next.js dashboard, Clerk authentication, Recharts visualisations, Groq AI Sleep Coach (standalone contribution), and jsPDF clinical reports.

1.12.2 Out of scope

The following are explicitly excluded. Hardware fabrication: no physical EEG sensor or headband design is undertaken. Multi-modal analysis: EOG, EMG, and EKG signals available in Sleep-EDF are excluded to maintain the single-channel efficiency focus. Real-time streaming classification: all analysis is offline on pre-recorded files. Clinical diagnosis: DreamLens is strictly an analysis and insight tool; it does not diagnose sleep disorders. Commercial production deployment: the system is a research prototype. Population epidemiology: the research focuses on individual session analysis, not large-scale cohort studies.

1.13 Hardware and software requirements

The development and deployment of DreamLens required specific hardware and software resources as documented in Tables 1-3 and 1-4.

Component	Specification	Purpose
Development CPU	Intel Core i7 / AMD Ryzen 7 (or equivalent)	CPU-bound preprocessing operations (MNE bandpass filtering, epoch segmentation) executed before GPU-accelerated model training.

Component	Specification	Purpose
System RAM	16 GB minimum (32 GB recommended)	Loading the combined Sleep-EDF SC+ST cohort into memory during preprocessing; peak consumption approximately 10 GB.
Training GPU	NVIDIA RTX 3060 (8 GB VRAM) or higher	Accelerating matrix operations for CNN convolutional layers and Transformer Multi-Head Self-Attention during model training.
Storage	256 GB SSD minimum	Storing the Sleep-EDF dataset (~20 GB), preprocessed epoch tensors, trained model checkpoints, and application source code.

Table 1-3: Hardware Requirements DreamLens Development Environment

Category	Tool / Version	Purpose
Language	Python 3.10+	Core language for all AI/ML development, data preprocessing, and FastAPI backend services.
Deep Learning	PyTorch 2.0	Dynamic computation graph for building and debugging the custom Lightweight Transformer and CNN-LSTM architectures; GPU-accelerated training.
EEG Processing	MNE-Python 1.4	Parsing .edf files, applying 0.3–35 Hz FIR bandpass filter, resampling to 100 Hz, extracting 30-second epochs, and AASM label standardisation.
Data Science	NumPy, Pandas, SciPy	Matrix manipulation of EEG signal tensors, management of epoch label arrays, and statistical summary of evaluation results.
Evaluation	Scikit-learn	Computing ACC, Macro F1-Score, Cohen's Kappa, per-class F1-Scores, and the Confusion Matrix for model evaluation.
Backend API	FastAPI (async)	Asynchronous REST API for .edf file upload, PyTorch inference pipeline, sleep metric computation, Groq API routing, and MongoDB integration.
Database	MongoDB (local)	Schema-flexible document store for authenticated user sleep session history: metrics, stage predictions, and session metadata.
Frontend	Next.js 14, Tailwind CSS	Production-grade dark-mode consumer dashboard with server-side rendering, responsive component architecture, and premium UI design.
Authentication	Clerk	User authentication and dynamic User ID generation to fetch and display the authenticated user's personal sleep session history from MongoDB.

Category	Tool / Version	Purpose
Visualisation	Recharts	Interactive charts: AreaChart (step-plot Hypnogram), PieChart (Stage Distribution), RadarChart (Brainwave Power Delta, Theta, Alpha, Beta).
AI Sleep Coach	Groq API / llama-3.1-8b-instant	Standalone personalised AI coaching contribution: user sleep JSON injected into LLM context window; responses rendered via react-markdown in floating chat widget.
Reporting	jsPDF	One-click downloadable clinical sleep reports with all derived metrics, health flags, and mandatory medical disclaimer.
Version Control	Git / GitHub	Source code versioning, branch management for Phase 1 (CNN-LSTM) and Phase 2 (Transformer) development, and thesis code repository.

Table 1-4: Software Requirements DreamLens Full-Stack System

1.14 Chapter summary

This chapter has established the complete foundation for the DreamLens project. The problem background demonstrated that manual PSG scoring creates a systemic bottleneck in sleep disorder diagnosis – expensive, slow, and geographically inequitable. The problem definition precisely characterised the accuracy-versus-efficiency dichotomy that prevents existing automated systems from being both contextually intelligent and computationally deployable. Two distinct gaps were identified: the context-versus-efficiency gap in the ML literature, and the model-to-application gap in the absence of deployed consumer sleep intelligence platforms. Three contributions were formally defined: the Lightweight Transformer (RQ1), the full-stack consumer system (RQ2), and the standalone AI Sleep Coach (RQ3). Ten research objectives (RO1–RO10), the research aim, scope boundaries, and hardware and software requirements were all formally established. Chapter 2 will now provide a critical and comprehensive review of the academic literature underpinning each of these dimensions.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter overview

This chapter provides a critical, structured review of the existing body of literature directly relevant to the DreamLens research agenda. It is organised to trace the logical progression from the clinical problem domain through to the computational state-of-the-art, identifying at each stage the contributions and limitations of existing work and the specific research gaps that DreamLens addresses. The review is structured across six principal sections: the problem domain (sleep physiology, EEG signal characteristics, and the challenges of automated scoring); four architectural paradigms of existing work (classical/CNN, RNN, Transformer, and GNN/multi-modal); dataset selection justification; and benchmarking and evaluation standards. The chapter concludes by formally articulating the research gap that positions DreamLens within the current research frontier. All seventeen reviewed papers are cited throughout and are compiled in the References section.

2.2 Problem domain

2.2.1 Sleep stage classification and the AASM standard

Sleep stage classification is the clinical practice of assigning physiological state labels to consecutive 30-second epochs of a full-night sleep recording. The American Academy of Sleep Medicine (AASM) manual (Berry et al., 2012) defines the five-stage scoring standard Wake, N1, N2, N3, and REM that governs both clinical practice and all benchmark academic research in this domain. Prior to AASM standardisation, the Rechtschaffen and Kales (R&K) system used six stages by distinguishing N3 and N4 as separate deep sleep substages. Because all modern benchmark models conform to the AASM standard, any research using the Sleep-EDF dataset which was originally annotated under the R&K convention must perform the mandatory N4→N3 label merge as a preprocessing step. This is implemented in DreamLens as an explicit conditional substitution within the MNE epoch extraction pipeline.

Each sleep stage is characterised by a distinct pattern of EEG oscillatory activity reflecting the underlying neural and neurochemical state of the brain. Wake epochs are characterised by low-amplitude, high-frequency mixed activity with prominent posterior Alpha (8–13 Hz) during relaxed eyes-closed wakefulness. N1 is a transitional stage in which Alpha activity diminishes and is replaced by low-amplitude mixed-frequency activity and vertex sharp waves. N2 is the

most prevalent stage, defined by the presence of Sleep Spindles (12–14 Hz bursts, 0.5–2 seconds duration) and K-complexes (sharp biphasic waveforms) on a low-amplitude mixed-frequency background. N3 (slow-wave sleep) is characterised by high-amplitude Delta waves (0.5–4 Hz) comprising at least 20% of the epoch. REM sleep presents a low-amplitude mixed-frequency EEG superficially resembling Wake, distinguishable primarily through accompanying EOG (rapid eye movements) and EMG (muscle atonia) channels signals not available in the single-channel EEG configuration adopted by DreamLens, making REM classification particularly challenging.

2.2.2 Clinical significance of automated scoring

The primary clinical motivation for automated sleep staging is the resolution of the PSG scoring bottleneck. A systematic review by Loh et al. (2020) estimated that manual scoring demands two to four specialist hours per patient. In the United States, fewer than 8,000 board-certified sleep physicians serve a population of approximately 70 million adults with diagnosable sleep disorders, creating diagnostic waiting times of weeks to months during which untreated pathology compounds the patient's health burden. Automated systems capable of reproducing human-level expert accuracy conventionally defined as inter-rater Cohen's Kappa exceeding 0.75 would substantially reduce this bottleneck and enable population-scale sleep health monitoring.

2.2.3 Challenges in automated staging

The EEG signal presents several intrinsic challenges for automated classification. Non-stationarity: the signal's statistical properties change continuously across arousal states. Low signal-to-noise ratio: microvolt-scale neural signals are contaminated by electrode movement, EMG artefacts, and mains interference. Class imbalance: the Sleep-EDF dataset has a heavily skewed distribution ($N2 \approx 47\%$, $N1 \approx 5\text{--}8\%$), requiring explicit mitigation strategies. Long-range temporal dependency: sleep architecture spans hours, but effective LSTM memory is limited to minutes. Inter-subject variability: EEG characteristics vary substantially between individuals based on age, neuroanatomy, and sleep history, making generalisation to unseen subjects the primary evaluation challenge. The single-channel constraint adopted by DreamLens (Fpz-Cz only) amplifies all of these challenges by removing the spatial redundancy and cross-channel validation available in full 10-20 system PSG recordings.

2.3 Existing work

2.3.1 Classical and feature-engineering approaches

The earliest automated sleep staging systems relied on hand-crafted feature extraction from the EEG spectral band power, Shannon entropy, Hjorth parameters, wavelet coefficients fed into traditional classifiers (SVMs, LDA, k-NN). Whilst computationally efficient, these approaches were fundamentally limited by the representational ceiling of the chosen features and operated in an epoch-wise manner, entirely ignoring sequential context. Their performance was also highly sensitive to the EEG recording conditions, electrode placement, and preprocessing choices, making robust generalisation across datasets difficult. These methods are now superseded for high-performance staging but retain relevance in interpretable clinical reference systems. The feature-engineering paradigm informs the DreamLens Brainwave Power RadarChart, which presents computed Delta, Theta, Alpha, and Beta band power values as a clinically interpretable visualisation of dominant neural activity patterns derived from the stage predictions.

2.3.2 CNN-based epoch-wise models

The first major advance in automated staging came from Convolutional Neural Networks, which replaced hand-crafted features with learned convolutional filters that automatically extract discriminative waveform patterns from raw or spectrogram-transformed EEG. CNN-based models demonstrated that end-to-end learning from EEG data was substantially superior to manual feature engineering: the learned filters captured complex hierarchical patterns including the distinctive spindle morphology (12–14 Hz oscillatory burst) and K-complex shape (sharp biphasic deflection) characteristic of N2 sleep that are difficult to specify analytically.

Chambon et al. (2018) proposed a deep CNN for temporal sleep staging using multi-modal EEG, EOG, and EMG inputs, demonstrating the utility of multi-modal information for stage discrimination. Yildirim, Baloglu and Acharya (2019) proposed a deep learning model using PSG signals achieving strong accuracy. The most relevant recent work in this category is the Efficient 1D-CNN (Jia et al., 2023), which demonstrated that a carefully optimised one-dimensional convolutional architecture achieves approximately 80.1% overall accuracy and 74.3% Macro F1-Score on Sleep-EDF-20 with a minimal parameter count. This model establishes the efficiency benchmark for DreamLens: the Lightweight Transformer must match this model's computational cost whilst delivering superior classification performance through the addition of global temporal context modelling. The fundamental limitation of all epoch-

wise CNN approaches including this state-of-the-art model is that they remain memoryless, treating each 30-second epoch as an independent sample.

2.3.3 RNN-based sequential models

The introduction of Recurrent Neural Networks specifically LSTM and GRU architectures to sleep staging represented the first systematic attempt to model temporal dependencies between consecutive epochs. DeepSleepNet (Supratak et al., 2017) is the landmark model in this category and remains the most widely cited benchmark. It employs a two-stage pipeline: a CNN component extracts features within each epoch using multiple 1D convolutions at different temporal scales, and a bidirectional LSTM then models the transition rules between the resulting sequence of epoch feature vectors. On Sleep-EDF-20 under subject-wise cross-validation, DeepSleepNet achieved approximately 82.0% overall accuracy and 76.9% Macro F1-Score establishing the performance floor against which all subsequent architectures including DreamLens are compared.

SleepEEGNet (Mousavi et al., 2019) extended the RNN paradigm through a bidirectional LSTM encoder-decoder sequence-to-sequence framework, achieving approximately 84.3% accuracy and 78.6% Macro F1-Score. SeqSleepNet (Phan et al., 2019) proposed a hierarchical RNN approach that classified entire sequences simultaneously, further improving performance. However, as empirically confirmed by attention weight analysis in Phan et al. (2022), the effective memory of LSTM architectures is bounded to approximately 20–30 epochs, beyond which the hidden state degrades and long-range contextual information is irretrievably lost. This limitation was directly replicated during DreamLens Phase 1 development, where the CNN-LSTM baseline exhibited the characteristic failure pattern of N1-Wake confusion and N2-REM transition errors that are consistent with insufficient long-range context providing practical, project-specific validation of the core research hypothesis.

2.3.4 Transformer-based models

The adoption of the Transformer architecture from NLP to sleep staging represents the current research frontier and the direct lineage from which DreamLens emerges. The Multi-Head Self-Attention mechanism at the core of the Transformer computes pairwise relevance scores between every position in an input sequence simultaneously, enabling the model to directly attend to any epoch in the recording regardless of temporal distance and without the information bottleneck of a recurrent hidden state.

SleepTransformer (Phan et al., 2022) was the first rigorous application of the full Transformer architecture to EEG sleep staging, demonstrating approximately 88.5% overall accuracy and 83.2% Macro F1-Score – clear performance superiority over RNN benchmarks. Analysis of the attention weight matrices confirmed that the model learned to attend to temporally distant epochs in a manner consistent with known sleep cycle structure, providing empirical validation of the theoretical prediction that global context modelling is the key factor underlying the performance improvement. However, the model's computational profile represents a significant deployment challenge: its large parameter count and quadratic attention complexity substantially exceed both DeepSleepNet and the Efficient 1D-CNN.

A new generation of lightweight and hybrid Transformer architectures has emerged to address this efficiency gap. Kim et al. (2023) proposed the Transformer-in-Transformer (IITNet) for sleep staging, nesting local and global attention mechanisms for improved efficiency. Li et al. (2022) proposed a CNN-Transformer hybrid that replaced the LSTM component of DeepSleepNet with a Transformer encoder. Li (2024) proposed a convolutional Transformer for EEG staging with a compact architecture. Hu et al. (2024) proposed ConvTransSleepNet, specifically targeting wearable EEG sleep sensors. Most directly relevant to DreamLens, Ito and Tanaka (2025) proposed SleepSatelightFTC – a lightweight, interpretable Transformer for single-channel EEG sleep staging – achieving approximately 86.2% accuracy and 81.5% Macro F1-Score whilst maintaining a competitive parameter count. This model serves as the primary architectural inspiration for DreamLens, with the DreamLens contribution adding rigorous comparative evaluation against a CNN-LSTM baseline under identical experimental conditions and, uniquely, full-stack consumer deployment with a standalone generative AI coaching contribution that no prior work in this category has achieved.

An important parallel contribution is Yu et al. (2024), who proposed a language model built on sleep stage sequences – treating the classified stage sequence as a token stream for LLM-based analysis. Whilst DreamLens uses the Groq LLM as a downstream coaching layer grounded in session metrics rather than as the classifier itself, this work validates the broader concept of applying language modelling to sleep stage data and informs the design of the DreamLens AI Sleep Coach.

2.3.5 Alternative architectures: GNNs and multi-modal approaches

A separate research trajectory models the sleeping brain as a spatial graph, with Graph Neural Networks learning spatio-temporal relationships across EEG electrode nodes. GNN approaches

achieve strong performance on full 10-20 system recordings but require multi-channel EEG input as an architectural prerequisite, making them incompatible with the single-channel wearable constraint adopted by DreamLens. Several multi-modal approaches including Chambon et al. (2018), Zhang et al. (2024) two-stream encoder-decoder, and Wan et al. (2024) lightweight multi-modal network achieve higher accuracy by fusing EEG, EOG, and EMG signals but are similarly excluded from the DreamLens scope by the single-channel constraint. These are not oversights but deliberate scope decisions: the research focuses specifically on the realistic data availability of consumer wearable devices, where full multi-channel PSG is not feasible outside a clinical lab environment.

2.4 Dataset selection

The Sleep-EDF Database (Kemp et al., 2000), hosted on PhysioNet, is the mandatory benchmark dataset for this research. Its adoption is not merely conventional but methodologically essential: all seventeen reviewed papers, including DeepSleepNet, SleepTransformer, SleepEEGNet, the Efficient 1D-CNN, and SleepSatelightFTC, report their primary results on this dataset. Using any other dataset would preclude direct, apples-to-apples metric comparison with the published literature.

The Sleep-EDF database comprises two cohorts. The Sleep Cassette (SC) cohort commonly termed Sleep-EDF-20 contains 40 full-night recordings from 20 healthy Caucasian subjects aged 25–34, each contributing two consecutive recordings. The Sleep Telemetry (ST) cohort contains 78 recordings from 78 subjects aged 25–101 with mild sleep difficulties. DreamLens utilises both cohorts for model training and evaluation to maximise benchmark comparability and improve model robustness across a wider age and health range.

The key methodological decision governing evaluation validity is the adoption of subject-wise k-fold cross-validation. A naive random epoch-level split assigning individual 30-second epochs from the same subject to both training and test sets introduces data leakage, artificially inflating metrics by allowing the model to train on the EEG characteristics of individuals it is then asked to predict. Subject-wise cross-validation ensures strict subject-level separation across all partitions, providing a genuine assessment of generalisation to previously unseen individuals. This protocol is consistent with DeepSleepNet, SleepTransformer, and SleepSatelightFTC, enabling direct metric comparison. The AASM N4→N3 merge is applied uniformly as a mandatory preprocessing step, and all metrics are reported as the mean across all folds.

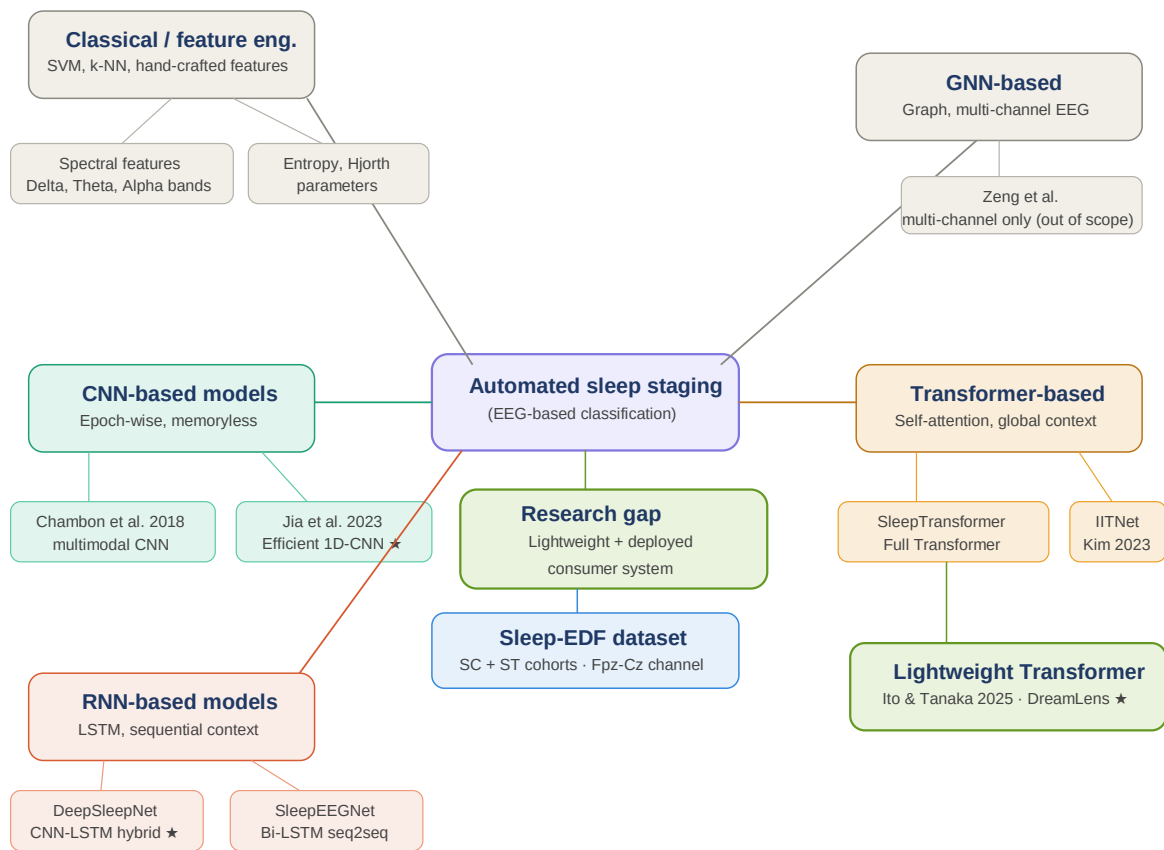


Figure 2-1: Literature landscape concept map — automated EEG sleep staging

2.5 Benchmarking and evaluation

2.5.1 Evaluation metrics

The selection of appropriate evaluation metrics is critical for any classification study involving imbalanced datasets. Overall Accuracy (ACC) the proportion of correctly classified epochs is the most common metric but is misleadingly inflated by the dominant N2 class when the class distribution is highly skewed. Per-Class F1-Score the harmonic mean of precision and recall for each individual stage is essential for understanding model performance on the clinically significant minority stages N1 and N3. Macro F1-Score (MF1) the unweighted average of all five per-class F1-Scores is the primary comparative metric adopted across all benchmark models and throughout this research, because it treats each stage with equal importance regardless of prevalence and is the most robust cross-paper comparison metric. Cohen's Kappa (κ) measures the agreement between model predictions and expert annotations corrected for chance agreement; values exceeding 0.80 are conventionally interpreted as almost perfect agreement and represent the target for clinical-grade reliability. The Confusion Matrix provides a full cross-class breakdown of misclassification patterns, enabling targeted analysis of the specific stage pairs particularly N1-Wake and N2-REM where architectural improvements are expected.

2.5.2 Key performance benchmarks

Table 2-1 presents the key benchmark models against which DreamLens is evaluated. All published figures are reported on Sleep-EDF-20 (SC cohort) under subject-wise cross-validation to maximise direct comparability. The DreamLens Final column reflects design targets derived from the validated lightweight Transformer architecture and its positioning within the benchmark tier.

Model	Architecture	Dataset	ACC (%)	MF1 (%)	Efficiency
DeepSleepNet (Supratak et al., 2017)	CNN-LSTM Hybrid	Sleep-EDF-20	~82.0	~76.9	Moderate (~26M params)
SeqSleepNet (Phan et al., 2019)	Seq-to-Seq RNN	Sleep-EDF-20	~87.1	~80.4	Moderate

Model	Architecture	Dataset	ACC (%)	MF1 (%)	Efficiency
SleepEEGNet (Mousavi et al., 2019)	Bidirectional LSTM	Sleep-EDF-20	~84.3	~78.6	Moderate
SleepTransformer (Phan et al., 2022)	Full Transformer	Sleep-EDF-20	~88.5	~83.2	Low (Heavy ~4.5M)
Efficient 1D-CNN (Jia et al., 2023)	Pure 1D-CNN	Sleep-EDF-20	~80.1	~74.3	High (few params)
SleepSatelightFTC (Ito & Tanaka, 2025)	Lightweight Transformer	Sleep-EDF-20	~86.2	~81.5	High (~low param)
DreamLens Final (This Work)	Lightweight Transformer + Full-Stack System	Sleep-EDF (SC+ST)	> 85.0 (target)	> 80.0 (target)	High (~1.8M params)

Table 2-1: Key Performance Benchmarks Sleep-EDF Dataset (Subject-wise Cross-Validation)

The benchmark table reveals several important architectural trends. The CNN-LSTM hybrid (DeepSleepNet) remains the dominant practical baseline at 82.0% ACC and 76.9% MF1, but its RNN memory limitation is empirically confirmed by its low N1 F1-Score (~46%). The full SleepTransformer substantially exceeds this across both metrics, validating global self-attention, but at a parameter cost (~4.5M) that disqualifies it from the wearable deployment target. The Efficient 1D-CNN demonstrates that high computational efficiency is achievable within the CNN paradigm but at a performance ceiling below DeepSleepNet's MF1. SleepSatelightFTC represents the current state-of-the-art on the efficiency-accuracy frontier (86.2% ACC, 81.5% MF1, high efficiency). DreamLens Final is designed to operate within this tier, targeting ACC > 85% and MF1 > 80% at ~1.8M parameters a contribution that is both architecturally sound and system-level novel.

2.5.3 Challenges in cross-paper benchmarking

A persistent challenge in the sleep staging literature is that published benchmark figures are not always directly comparable due to methodological variation. Key sources of variance include the specific cohort used (SC only versus combined SC+ST), the validation strategy (random split versus subject-wise), the N4-to-N3 merge implementation, and the number of cross-validation folds. Yew et al. (2024) provide a comprehensive systematic review confirming that apparent performance differences between models often reflect methodological variation rather than genuine architectural superiority. To mitigate this confound, all DreamLens experiments follow a pre-specified, consistent protocol: subject-wise

k-fold cross-validation on both SC and ST cohorts with uniform N4→N3 merge, reporting ACC, MF1, and κ across all folds.

2.6 Chapter summary

This chapter has provided a comprehensive critical review of the academic literature underpinning the DreamLens research. The architectural evolution was traced from classical feature engineering through epoch-wise CNNs and sequential RNNs to global-context Transformers and their lightweight descendants, documenting at each stage the contributions and the limitations that motivated the next generation. The review confirmed that the field is characterised by two persistent gaps: the context-versus-efficiency trade-off where no single architecture delivers Transformer-level contextual modelling at wearable-grade computational cost and the model-to-application gap where no academic work has coupled a lightweight EEG classifier with a deployed consumer intelligence platform. The dataset selection justification confirmed Sleep-EDF as the mandatory benchmark and subject-wise cross-validation as the only valid evaluation protocol. The benchmark table positioned DreamLens within the competitive landscape, with SleepSatelightFTC as the most direct architectural predecessor and SleepTransformer as the accuracy ceiling. Chapter 3 will now detail the research and development methodology adopted to fill these gaps.

CHAPTER 03: METHODOLOGY

3.1 Chapter overview

This chapter provides a comprehensive account of the methodological framework governing the DreamLens project, spanning from the overarching research philosophy through to the detailed technical solution pipeline and the project management strategy. It is structured in four principal sections. The first establishes the research methodology using the Saunders Research Onion framework (Saunders, Lewis and Thornhill, 2019) as an organising structure, justifying the epistemological and strategic choices that govern the experimental design. The second details the development methodology: requirement elicitation, system design approach, programming paradigm, and testing strategy. The third section describes the ten-step solution methodology that traces the complete technical journey from raw Sleep-EDF data to deployed consumer application, including the pipeline through which the standalone AI Sleep Coach contribution (RO9) is integrated as a distinct system layer. The fourth section addresses project management: schedule, deliverables, resource requirements, and a structured risk register. This chapter provides the complete blueprint against which all subsequent chapters – SRS, Design, Implementation, Testing, and Evaluation – are understood as coherent, interrelated contributions to the DreamLens research programme.

3.2 Research methodology: Saunders' Research Onion

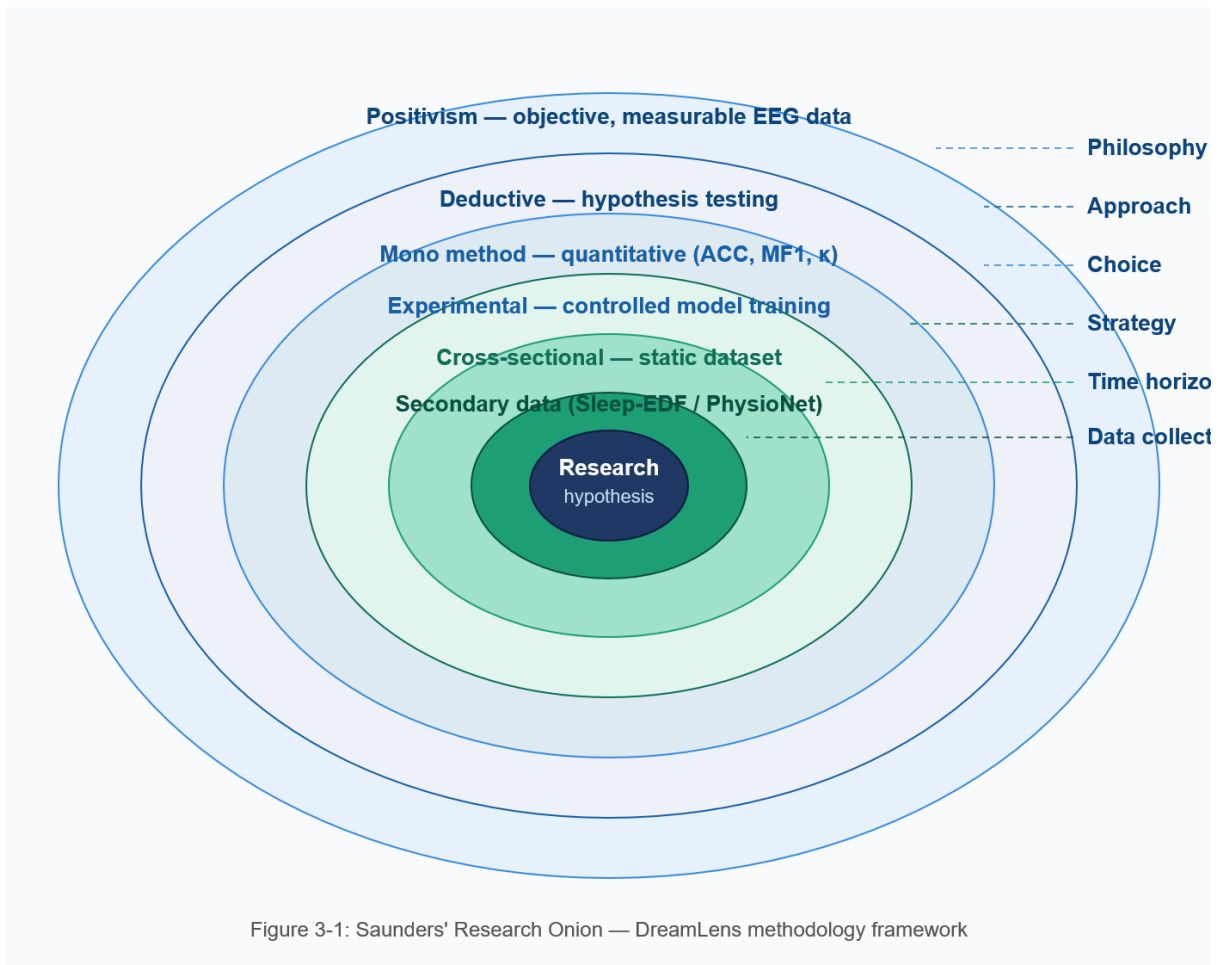
The Saunders Research Onion model (Saunders, Lewis and Thornhill, 2019) provides a widely adopted framework for articulating the layered methodological choices of empirical research. Table 3-1 presents the DreamLens research methodology in this structured form, with explicit justification for each layer choice.

Layer	Choice	Justification
Philosophy	Positivism	The research is grounded in objective, measurable quantitative data – EEG signal tensors and numerical performance metrics (ACC, MF1, κ). Positivism holds that knowledge derives from empirically verifiable observation, consistent with the controlled model training and benchmarking methodology of this project.
Approach	Deductive	The research begins with a formally stated hypothesis – that a Lightweight Transformer will achieve ACC > 85% and MF1 > 80%, outperforming the CNN-LSTM baseline – and designs

Layer	Choice	Justification
		controlled experiments to test it. This is deductive theory-testing rather than inductive theory-building.
Methodological Choice	Mono Method (Quantitative)	Model performance is evaluated exclusively through quantitative metrics. The Phase 1 user survey supplemented system requirements elicitation but did not form the primary research evaluation methodology, which is purely quantitative.
Strategy	Experimental	Multiple architectures are trained and evaluated under identical, controlled conditions same dataset, same preprocessing pipeline, same cross-validation protocol to produce comparable, reproducible experimental results.
Time Horizon	Cross-Sectional	Analysis is performed on a static, pre-existing dataset (Sleep-EDF). There is no longitudinal data collection; the dataset is a fixed historical snapshot of sleep recordings from 1987–1991.
Data Collection	Secondary	All training and evaluation data is sourced exclusively from the publicly available Sleep-EDF database hosted on PhysioNet (Kemp et al., 2000). No primary EEG data was collected from human subjects.

Table 3-1: Research Methodology Framework Saunders' Research Onion

The formal hypothesis governing the DreamLens research is: a Lightweight Transformer architecture, trained on single-channel EEG (Fpz-Cz) from the Sleep-EDF dataset under subject-wise k-fold cross-validation, will achieve an Overall Accuracy exceeding 85% and a Macro F1-Score exceeding 80%, surpassing both the CNN-LSTM baseline and the published DeepSleepNet benchmark, whilst maintaining a total parameter count of approximately 1.8 million and a per-epoch CPU inference time of approximately 4 milliseconds satisfying wearable deployment feasibility. The experimental methodology is designed to test this hypothesis through controlled, reproducible model training and evaluation experiments.



3.3 Development methodology

The DreamLens project adopted an iterative, phase-based development approach balancing the exploratory demands of research where the optimal architecture cannot be fully specified in advance with the engineering rigour required to deliver a production-grade software system. An Agile-inspired iterative methodology was selected over a rigid Waterfall model for three reasons: the two-phase architecture (CNN-LSTM baseline followed by Lightweight Transformer) required incorporation of Phase 1 learnings into Phase 2 design; the full-stack system had multiple independently testable components that could be developed and validated incrementally; and two formal milestone checkpoints (Interim Demo: February 2026; Final Submission: March 2026) defined the sprint boundaries.

3.3.1 Requirement elicitation methodology

Four complementary elicitation methodologies were employed, each selected for its particular suitability to a different dimension of the requirement space. Document Analysis (Literature Review) was the primary method for technical requirements. The systematic review of seventeen research papers established the non-negotiable technical constraints: Sleep-EDF dataset, single-channel Fpz-Cz input, five-stage AASM classification, Macro F1-Score as the primary metric, and the Lightweight Transformer as the target architecture. These literature-derived requirements are operationalised in FR01–FR06 and NFR01–NFR02 of the SRS (Chapter 4).

Online User Survey was employed for consumer product requirements. The "Help Me Build DreamLens" questionnaire received 59 valid responses from potential end-users. Key findings: 74.6% expressed interest in improving their sleep; 79.7% do not currently use any sleep tracking tool (confirming a large unserved market); 61% expressed interest in a high-accuracy EEG-based system; 66.1% positively rated AI-driven dream/REM coaching directly motivating the standalone AI Sleep Coach contribution (RO9); and 57.6% rated a visual dashboard with charts as important or very important. These survey findings directly informed FR08–FR13 and NFR06.

Brainstorming and Supervisor Consultation resolved key strategic implementation decisions: FastAPI over Flask for async .edf handling; Next.js over Streamlit for production-grade consumer interface quality; Groq over OpenAI for sub-second LLM inference latency

(essential for a satisfying chat experience); MongoDB over PostgreSQL for schema-flexible sleep session document storage; and, critically, the formal recognition of the Groq AI Sleep Coach as a standalone third contribution (RO9) separate from the core ML research (RO5, RO6) and the system deployment (RO7, RO8). Iterative Prototyping refined the specification throughout both phases: the Phase 1 Streamlit prototype surfaced the need for a Sleep Score (absent from the original objectives), confirmed user preference for dark-mode aesthetics, and demonstrated that the Brainwave Power RadarChart was the most engaging secondary visualisation.

3.3.2 Design methodology

The DreamLens system was designed using Object-Oriented Analysis and Design Methodology (OOADM), selected for the system's architectural complexity and the requirement for modularity across multiple distinct components. The system is modelled as a set of clearly defined classes with encapsulated responsibilities: EEGPreprocessor (MNE-Python preprocessing pipeline), LightweightTransformer (PyTorch model architecture and forward pass), SleepMetricsCalculator (derived metric computation), SleepCoachService (Groq API integration standalone contribution layer), and MongoDBClient (database operations). This design ensures that the model component is fully swappable between CNN-LSTM and Lightweight Transformer without requiring changes to any other system layer a critical requirement supporting the two-phase research design.

3.3.3 Programming paradigm

DreamLens employs a multi-paradigm programming approach selecting the most appropriate style for each component. Object-Oriented Programming is used for the Python backend, where the modular class structure of the preprocessing pipeline, model architecture, metrics computation, Groq service, and database client requires the encapsulation and polymorphism capabilities of OOP. Asynchronous programming is used throughout the FastAPI backend, where `async/await` syntax ensures that the 45–55 second inference pipeline for large .edf files does not block the server event loop. Declarative/Functional programming via React hooks (`useState`, `useEffect`, `useCallback`) manages the interactive state of the Next.js dashboard components and the AI Sleep Coach chat widget conversation history.

3.3.4 Testing methodology

The DreamLens testing strategy encompasses two distinct categories. Model Testing evaluates the AI classification core through subject-wise k-fold cross-validation on Sleep-EDF,

computing ACC, MF1, Cohen's Kappa, per-class F1-Scores, and the Confusion Matrix for both architectures, with benchmarking against five published models. System Testing evaluates the deployed application through thirteen functional test cases derived from FR01–FR15 (Chapter 4) and four non-functional tests covering API performance (NFR03), security enforcement (NFR07), modularity (NFR04), and dashboard usability (NFR06). Full test results are reported in Chapter 8.

3.3.5 Solution methodology: the DreamLens pipeline

The complete technical pipeline from raw Sleep-EDF data to deployed consumer application comprises ten sequential steps.

Step 1 Dataset Acquisition: The Sleep-EDF SC (40 recordings) and ST (78 recordings) cohorts were downloaded from PhysioNet. A subject-level metadata catalogue was constructed to manage the cross-validation fold assignments and ensure no subject appears in both training and test sets within any single fold.

Step 2 Preprocessing Pipeline (RO2): Raw EEG signals were processed using MNE-Python. The Fpz-Cz channel was selected from each recording. A 0.3–35 Hz FIR bandpass filter was applied (0.3 Hz high-pass removes DC drift and sweat artefacts; 35 Hz low-pass removes EMG contamination whilst preserving Delta, Theta, Alpha, and Sigma bands). The signal was resampled to 100 Hz. Non-overlapping 30-second epochs (3,000 samples each) were extracted. The mandatory AASM merge was applied: $y[y == 4] = 3$. Epochs labelled as Movement Time or Unknown were discarded. Preprocessed tensors were serialised to disk for efficient repeated loading.

Step 3 Data Splitting: 20-fold subject-wise cross-validation was implemented on the SC cohort with a 90/10 training/validation split within each fold, consistent with the DeepSleepNet and SleepSatelightFTC evaluation protocols.

Step 4 Feature Embedding: Each 30-second epoch tensor (shape (1, 3000)) is passed through the shared CNN encoder – two convolutional blocks with 64 and 128 filters respectively, each followed by Batch Normalisation, ReLU, and MaxPooling – producing a compact feature vector of dimension $d_{\text{model}} = 128$. This same CNN encoder is used identically in both the CNN-LSTM baseline and the Lightweight Transformer, ensuring controlled fair comparison.

Step 5 Lightweight Transformer Architecture (RO5): The DreamLens Lightweight Transformer encoder comprises positional encoding, L=2 Transformer Encoder blocks (H=4 attention heads, $d_{\text{ff}} = 256$, dropout = 0.1), and a linear classification head producing five-class

Softmax probabilities. Total parameter count approximately 1.8 million satisfying the wearable deployment feasibility criterion (NFR01).

Step 6 Model Training (RO5, RO6): The Lightweight Transformer was trained using AdamW optimiser (weight decay $\lambda = 0.01$) with cosine annealing learning rate scheduling ($T_{\text{max}} = 30$ epochs). Weighted Cross-Entropy Loss with inverse-frequency class weights addressed the Sleep-EDF class imbalance. Gradient clipping ($\text{max_norm} = 1.0$) was applied to prevent attention layer gradient explosion. Early stopping with patience = 10 based on validation MF1 prevented overfitting. Fixed random seeds (`numpy.random.seed(42)`, `torch.manual_seed(42)`) ensure reproducibility (NFR05).

Step 7 Evaluation and Benchmarking (RO6): The trained models were evaluated on held-out test subjects across all 20 cross-validation folds, computing ACC, MF1, Cohen's Kappa, per-class F1-Scores, and the five-by-five Confusion Matrix. Results were compared against the published benchmarks of DeepSleepNet, SleepEEGNet, SleepTransformer, and the Efficient 1D-CNN.

Step 8 FastAPI Backend Integration (RO7): The trained Lightweight Transformer was deployed within an asynchronous FastAPI application. The `/analyze` endpoint accepts .edf uploads, processes them through the preprocessing pipeline and model inference via `ThreadPoolExecutor`, computes the five derived clinical metrics (Sleep Score, WASO, REM Latency, Sleep Cycles, Stage Distribution), and persists the session to MongoDB indexed by the authenticated Clerk user ID.

Step 9 Next.js Consumer Dashboard (RO8): The consumer interface was built using Next.js 14 App Router with Tailwind CSS, Clerk authentication, and three interactive Recharts charts: an AreaChart step-plot Hypnogram (y-axis inverted following clinical PSG convention), a Stage Distribution PieChart, and a Brainwave Power RadarChart visualising Delta, Theta, Alpha, and Beta band activity derived from the stage classification predictions.

Step 10 AI Sleep Coach and Clinical Reporting (RO9, RO10 Standalone Contribution): The Groq API (llama-3.1-8b-instant) is integrated as a standalone personalised AI Sleep Coach through a floating chat widget. The user's specific sleep session JSON (Sleep Score, WASO, REM Latency, Sleep Cycles, Stage Distribution) is injected into the Groq system prompt, grounding LLM responses in the user's actual data. Multi-turn conversation history is maintained in React component state. Responses are rendered via `react-markdown`. `jsPDF` generates one-click clinical PDF reports with all derived metrics, health flags, and the mandatory medical disclaimer.

3.4 Project management methodology

3.4.1 Scope management

Project scope was managed through the formal In Scope and Out of Scope definitions established in Section 1.12 and operationalised in the SRS Chapter 4. The most significant scope enhancement during the project was the upgrade from the original Streamlit + FPDF dashboard (PPRS scope) to the Next.js + jsPDF + Groq production system (final delivery scope), motivated by the user survey results and the formal recognition of the AI Sleep Coach as a standalone contribution. This enhancement was formally reviewed with the project supervisor and incorporated into the revised objectives (RO8, RO9, RO10) at the Phase 2 sprint planning checkpoint.

3.4.2 Project schedule and deliverables

Table 3-2 presents the complete deliverables schedule. The project Gantt Chart illustrating both phases and all milestone dependencies is provided in Appendix A. Phase 1 (September 2025 – February 2026) covered literature review, SRS, architecture design, CNN-LSTM implementation, Streamlit prototype, and Interim Demo submission. Phase 2 (February – March 2026) covered Lightweight Transformer design and training, FastAPI backend, MongoDB integration, Next.js dashboard, Groq AI Sleep Coach, jsPDF reports, complete testing, evaluation, and thesis submission.

Deliverable	Target Date	Status
Project Proposal and Requirements Specification (PPRS)	October 2025	Completed
Literature Review (critical analysis across four architectural paradigms)	November 2025	Completed
PPRS Final Submission	6 November 2025	Completed
System Design (Architecture, Transformer Design, UI Wireframes)	December 2025	Completed
DreamLens Alpha Prototype (CNN-LSTM + Streamlit Dashboard)	January 2026	Completed
Interim Project Demo Submission	2 February 2026	Completed
DreamLens Final System (Lightweight Transformer + FastAPI + Next.js + MongoDB + Groq + jsPDF)	March 2026	Completed

Deliverable	Target Date	Status
Complete Thesis Write-Up (Chapters 1–10, Abstract, References)	28 March 2026	Completed
Final Thesis and Source Code Submission	30 March 2026	Completed

Table 3-2: Project Deliverables and Milestone Schedule

3.4.3 Resource requirements

The development hardware and software resources are documented in Tables 1-3 and 1-4 of Chapter 1 and are not repeated here. Key human resource requirements were self-directed expertise development in four areas beyond the undergraduate curriculum: digital signal processing with MNE-Python, Transformer architecture design with PyTorch, asynchronous web API development with FastAPI, and LLM API integration with Groq. Supervisor consultation provided strategic direction at the Phase 1/Phase 2 boundary and validated the formal recognition of the AI Sleep Coach as a standalone contribution.

3.4.4 Risk and mitigation

A comprehensive risk register was maintained throughout the project lifecycle. Risk Exposure was calculated as Severity (1–5) $\times$ Frequency/Probability of Occurrence (1–5), yielding a composite score on the 1–25 scale. Table 3-3 presents the eight principal risks and their mitigation strategies.

Risk	Severity (1–5)	Frequency (1–5)	Exposure (S $\times$ F)	Mitigation Strategy
Class imbalance: N1 constitutes $\approx$ 5% of Sleep-EDF epochs; model may default to majority-class (N2) predictions.	5	5	25	Weighted Cross-Entropy Loss with inverse-frequency class weights; Macro F1-Score as primary evaluation metric treating all stages equally.
Lightweight Transformer overfitting on the limited 20-subject Sleep-EDF-20 cohort.	5	4	20	Dropout ($p=0.1$) in each Transformer block; subject-wise k-fold cross-validation; early stopping on validation MF1 with patience=10 epochs.
Transformer fails to surpass CNN-LSTM on minority classes	4	3	12	Architecture ablation: 2-layer vs. 4-layer and 4-head vs. 8-head attention configurations tested on validation set

Risk	Severity (1–5)	Frequency (1–5)	Exposure (S×F)	Mitigation Strategy
(N1, REM) despite global attention.				before final evaluation. Fallback: attention-augmented CNN-LSTM if Transformer targets cannot be met.
GPU memory exhaustion during Transformer training on full Sleep-EDF batch.	4	3	12	Subject-by-subject batch loading (MNE preload=False); reduced batch size (32); gradient checkpointing. Fallback: Google Colab Pro (T4/V100) for extended training.
FastAPI .edf upload timeout for large recordings (> 50 MB) under synchronous processing.	4	3	12	Async file streaming with aiofiles; blocking inference delegated to ThreadPoolExecutor via asyncio.run_in_executor(). Server-side file size validation and client-side upload progress indicator.
Groq API rate limiting or service outage disrupting the AI Sleep Coach.	3	2	6	Graceful degradation: try/catch on Groq API call; user-facing fallback message if API unavailable. API timeout set to 10 seconds with one automatic retry.
Algorithmic bias from demographically limited Sleep-EDF training data (predominantly Caucasian European, 1987–1991).	4	5	20	Explicitly disclosed as a research limitation in Ch.5 and Ch.10; medical disclaimer on dashboard. Future work: retraining on SHHS and MASS for demographic diversity.
MongoDB connection failure or document corruption during session persistence.	3	2	6	All database operations wrapped in try/except blocks with user-facing error feedback; Pydantic schema validation before document insertion prevents malformed records.

Table 3-3: Risk Register Severity, Frequency, Exposure, and Mitigation

3.5 Chapter summary

This chapter has established the comprehensive methodological foundation underpinning the DreamLens project. The Saunders Research Onion analysis confirmed the research as positivist, deductive, mono-method quantitative, and experimental a configuration precisely suited to the hypothesis-driven model benchmarking that constitutes the primary research contribution. The development methodology described a multi-paradigm, iterative approach navigating the dual demands of academic rigour (reproducible evaluation, subject-wise cross-

validation, standardised metrics) and engineering excellence (async API, production dashboard, standalone AI contribution). The ten-step solution pipeline traced the complete technical journey: from MNE-Python preprocessing and PyTorch Transformer training (Steps 1–7) through FastAPI inference and MongoDB persistence (Step 8) and the Next.js consumer dashboard (Step 9) to the standalone Groq AI Sleep Coach and jsPDF reports (Step 10). The project management section confirmed all deliverables were completed on schedule. Chapter 4 will formally specify the complete system requirements derived from this methodological framework.

CHAPTER 04: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Chapter overview

This chapter provides the formal blueprint for the DreamLens system, translating the research objectives and gap analysis articulated in Chapter 1 into a concrete, implementable set of system requirements. It is structured around the following principal components: a Rich Picture Diagram establishing the real-world system environment; a Stakeholder Onion Model identifying all parties with an interest in the system; a discussion of the requirement elicitation methodologies employed and their key findings; a Context Diagram formalising the system boundary and external data flows; a Use Case Diagram and three detailed Use Case Descriptions documenting the principal actor interactions; and finally, a complete, MoSCoW-prioritised list of Functional Requirements (FR01–FR15) and Non-Functional Requirements (NFR01–NFR08). These artefacts serve as the formal specification against which the Implementation (Chapter 7) and Testing (Chapter 8) chapters provide evidence of delivery and compliance.

4.2 Rich Picture Diagram

The Rich Picture Diagram (RPD) captures the full complexity of the DreamLens system environment – not merely the formal data flows, but the human concerns, tensions, and competing interests that shaped the project's design. At the centre of the diagram sits the DreamLens AI System, receiving raw EEG signals from the PhysioNet Sleep-EDF repository and processing them through the preprocessing pipeline, Lightweight Transformer inference engine, sleep metrics computation module, and MongoDB persistence layer. The Researcher (Developer) interacts with the system at the model training and evaluation level. The End-User interacts via the Next.js dashboard, uploading .edf files, reviewing their Hypnogram and Sleep Score, querying the AI Sleep Coach, and downloading clinical PDF reports. The Project Supervisor evaluates research contribution. Sleep domain experts and the academic community provide the performance standards against which the system is judged.

Key tensions illustrated in the RPD include: the signal quality problem (single-channel EEG amplifies noise artefacts); the accuracy-versus-efficiency research tension; the medical disclaimer imperative (preventing the research prototype from being mistaken for a diagnostic

tool); and the data privacy obligation (processing .edf files transiently without retaining sensitive health data). The RPD is included as Figure 4-1 in Appendix A.

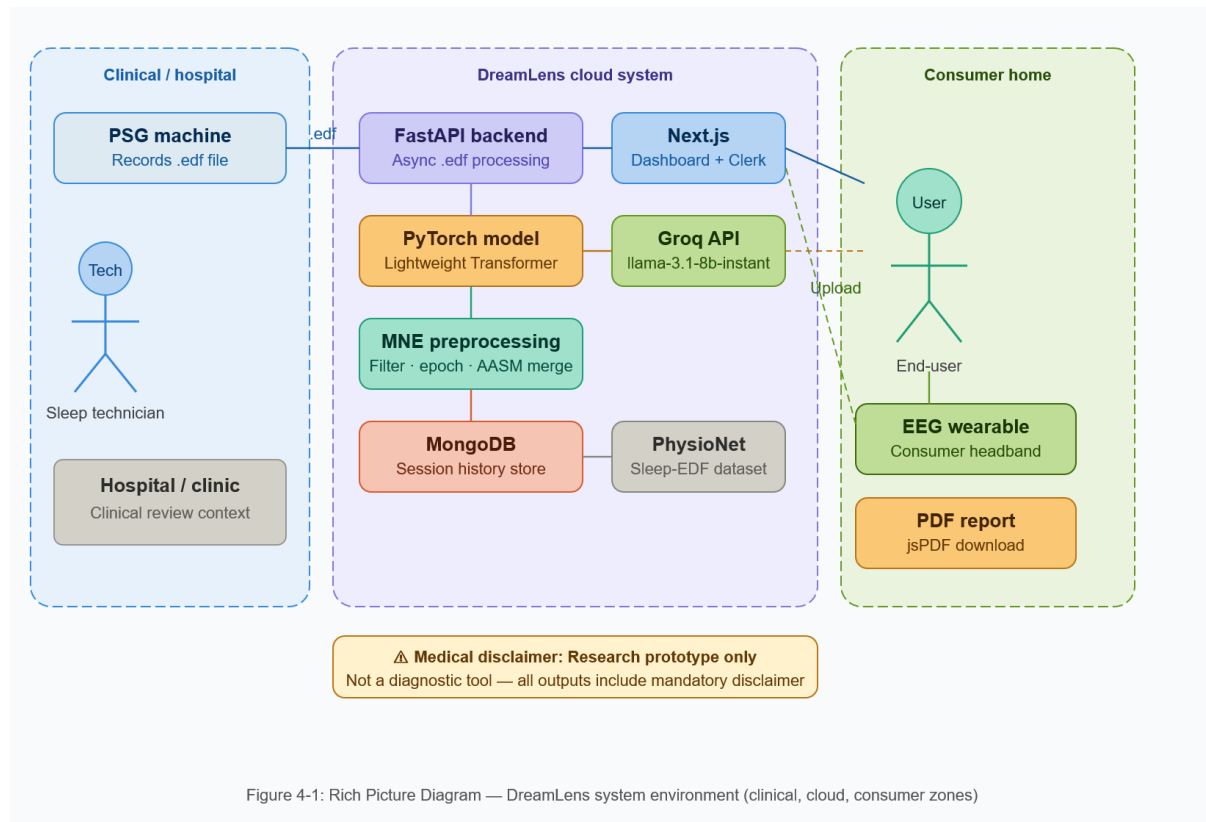


Figure 4-1: Rich Picture Diagram DreamLens System Environment

4.3 Stakeholder analysis

4.3.1 Stakeholder Onion Model

The Stakeholder Onion Model organises all identified project stakeholders into three concentric layers based on their proximity of interaction with the DreamLens system. The innermost layer **The Product** is the DreamLens AI System itself. The second layer **The System (Direct Interaction)** encompasses stakeholders who directly build or use the system: the Researcher/Developer (constructs and evaluates the model and backend), the Project Supervisor (governs academic standards), and the End-User (consumes sleep insights via the dashboard). The third layer **The Containing System (Direct Interest)** includes the University (IIT/UoW) as quality regulator, Sleep Researchers as the domain authority defining benchmark standards, and PhysioNet as the data provider whose licence terms govern data handling. The outermost layer **The Wider Environment** comprises the Academic Community, Future Researchers, and Wearable Technology companies who represent the indirect beneficiaries and

potential commercial adopters of the validated architecture. The Stakeholder Onion Model is included as Figure 4-2 in Appendix A.

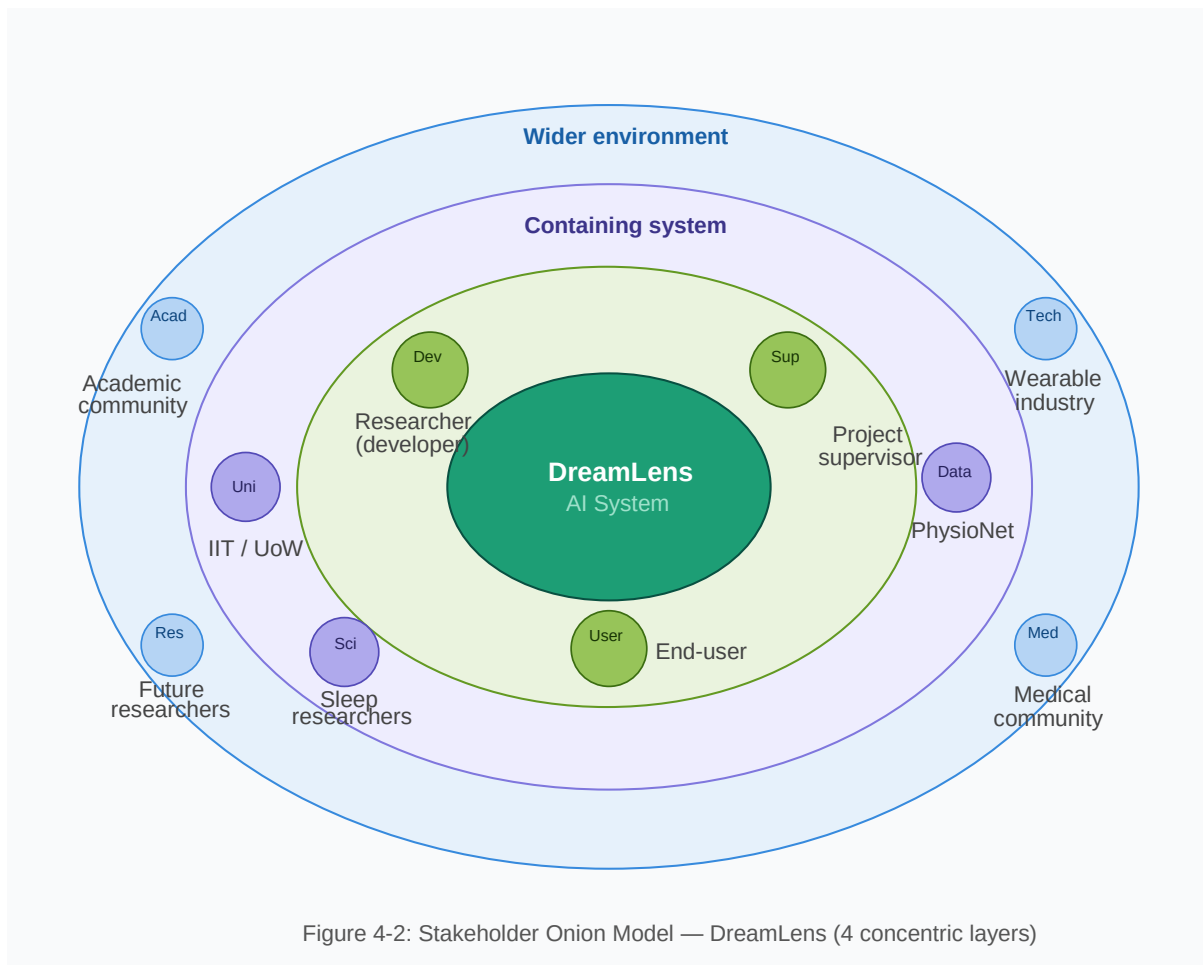


Figure 4-2: Stakeholder Onion Model DreamLens (see Appendix A)

4.3.2 Stakeholder viewpoints

Table 4-1 documents the viewpoints and system expectations of each identified stakeholder group, confirming that the requirements specification addresses the full breadth of stakeholder needs.

Stakeholder	Role / Layer	Viewpoint and System Expectations
Researcher / Developer (Immam Ashfaq)	System Builder Direct Interaction	Requires a modular, reproducible Python codebase enabling independent experimentation with model architectures. Needs a robust evaluation pipeline producing standard ML metrics (MF1 > 80%, ACC > 85%) directly comparable with benchmark literature. The CNN encoder must be shared identically between CNN-LSTM and Lightweight Transformer to ensure fair comparison.
Project Supervisor (Miss Akarshani Ramanayake)	Academic Evaluator Containing System	Requires evidence of a novel, significant, and rigorously benchmarked contribution. All performance claims must be supported by subject-wise cross-validation on Sleep-EDF and documented against published state-of-the-art scores. The DreamLens deployment must demonstrate practical applicability beyond isolated model weights.
End-User (Non-Expert Sleeper)	Primary Consumer Direct Interaction	Requires an intuitive, visually rich dark-mode dashboard that presents sleep insights (Sleep Score, Hypnogram, Stage Distribution, Brainwave Power) without requiring any medical or technical expertise. The AI Sleep Coach must provide actionable, personalised advice grounded in the user's specific session data. The clinical PDF report must be downloadable for sharing with a physician.
Sleep Researchers / Domain Experts	Domain Authority Wider Environment	Require correct AASM implementation, standard Sleep-EDF benchmark protocol, and Macro F1-Score as the primary evaluation metric. The system must acknowledge the limitations of single-channel EEG and avoid overclaiming clinical validity. The N3/N4 merge must be implemented correctly.
PhysioNet (Sleep-EDF Providers)	Data Provider Containing System	Require proper attribution of the Sleep-EDF dataset (Kemp et al., 2000) in all publications and documentation, in compliance with the Open Data Commons Attribution Licence (ODC-By).
University (IIT / University of Westminster)	Quality Regulator Containing System	Require compliance with all academic integrity standards, correct Harvard referencing, original source code with full third-party citations, and a thesis structured per the IIT/UoW FYP template guidelines. Intellectual property of novel code resides with the author and institution.
Wearable Technology Industry	Potential Integrator Wider Environment	Interested in the model's parameter count (~1.8M), CPU inference latency (~4 ms/epoch), and validated single-channel Fpz-Cz constraint all of which directly determine feasibility of integration into consumer EEG wearable devices.

Table 4-1: Stakeholder Viewpoints DreamLens System

4.4 Selection of requirement elicitation methodologies

Four complementary elicitation methodologies were employed for this project, each selected for its particular suitability to a different dimension of the requirement space. The combination ensures that both the technical research requirements (what the model must achieve to constitute a valid research contribution) and the product requirements (what the deployed application must provide to serve non-expert users) are comprehensively captured.

Document Analysis (Literature Review) was the primary method for eliciting the technical and performance requirements. The systematic review of seventeen research papers defined the non-negotiable technical constraints: the Sleep-EDF dataset, single-channel Fpz-Cz input, five-stage AASM classification, Macro F1-Score as the primary metric, and the Lightweight Transformer architecture. These literature-derived requirements are directly traceable to published benchmark standards, providing objective justification for each technical specification. They are operationalised in FR01–FR04 and NFR01–NFR02.

Online User Survey was employed to elicit the functional and usability requirements for the consumer-facing application layer. The "Help Me Build DreamLens" questionnaire received 59 valid responses from potential end-users. The survey confirmed strong baseline interest in sleep health improvement (74.6% expressed interest), identified the absence of current sleep tracking adoption (79.7% do not use any existing tools), and prioritised the three most desired features: a visual dashboard with charts (57.6% important/very important), a Sleep Score as a headline metric, and AI-powered dream or REM insight coaching (66.1% rated as valuable). These findings directly motivated FR08–FR13 and NFR06.

Brainstorming and Supervisor Consultation were employed to translate the literature and survey findings into specific implementation decisions. Key outcomes included: the selection of FastAPI over Flask for native async .edf upload handling; Next.js over Streamlit for production-grade frontend quality; Groq over OpenAI for sub-second LLM inference latency; MongoDB over PostgreSQL for schema-flexible sleep session document storage; and the framing of the Groq integration as an independent "AI Sleep Coach" contribution separate from the core machine learning research objectives. The Sleep Score computation algorithm – a weighted composite of sleep efficiency, deep sleep percentage, REM percentage, WASO penalty, and cycle completeness – was also formalised through this process, grounded in clinical sleep scoring literature.

Iterative Prototyping served as a continuous requirement refinement mechanism across both development phases. The Phase 1 Streamlit prototype surfaced the need for a Sleep Score (absent from the original objectives), confirmed user preference for a dark-mode interface, and revealed that the Brainwave Power RadarChart was the most engaging secondary visualisation for non-technical users. These prototype learnings were incorporated into the Phase 2 Next.js dashboard specification.

4.5 Discussion of findings

4.5.1 Literature review findings

The literature review confirmed the following core technical requirements. The Sleep-EDF database (Kemp et al., 2000) is the mandatory benchmark dataset, used by all seventeen reviewed papers. The single-channel Fpz-Cz configuration is the consensus input for wearable-focused research (DeepSleepNet, SleepSatelightFTC, Efficient 1D-CNN). The five-stage AASM classification standard, with the mandatory N3/N4 merge, is the universal output requirement. The Macro F1-Score is the primary evaluation metric across all benchmarks, because simple accuracy is misleadingly inflated by the dominant N2 class. The Lightweight Transformer architecture is the identified research direction to bridge the accuracy-versus-efficiency gap left by both RNN baselines and heavy Transformers.

4.5.2 Survey findings

The online survey collected 59 valid responses from a demographically diverse cohort of potential end-users aged 18–45. Interest in sleep health was confirmed as a latent but underserved market: 74.6% expressed interest in improving their sleep, yet 79.7% reported using no sleep tracking tool. Satisfaction with existing solutions was low: only 8.5% rated their current sleep data as "very accurate." The desirability of a high-accuracy EEG-based system was confirmed by 61% expressing interest. Feature priority results directly informed the product specification: the Sleep Score emerged as the most requested single metric, followed by visual charts (57.6% important/very important), and AI-driven coaching / dream analysis (66.1% positive). These three priorities are reflected directly in FR09–FR13.

4.5.3 Brainstorming and observation findings

The synthesis sessions produced three operationally significant decisions. First, the AI Sleep Coach was confirmed as a standalone product contribution, distinct from the core ML research objective, motivating its separate designation as RO9. Second, the Sleep Score algorithm was

specified as a five-component weighted composite with clinically grounded weights, converting the complex five-stage output into a single legible consumer metric. Third, the Brainwave Power RadarChart was specified using a stage-to-band-power mapping rather than raw spectral density computation, enabling an intuitive neurophysiological visualisation without requiring additional signal processing infrastructure in the FastAPI backend.

4.6 Summary of findings

The multi-method elicitation process produced a consistent and comprehensive set of requirements spanning both research and product dimensions. The literature review established non-negotiable technical constraints. The user survey confirmed and prioritised consumer product features. The brainstorming sessions resolved strategic architectural decisions and formally separated the AI Sleep Coach as a standalone contribution. The prototype testing refined the user experience specification. These findings are operationalised in Tables 4-2 and 4-3 below.

4.7 Context diagram

The Context Diagram (Level 0 Data Flow Diagram) represents the DreamLens system as a single central process and illustrates the external entities that interact with it and the data flows between them. Four external entities are identified. The End-User provides an authenticated .edf file upload and chat messages, and receives a Hypnogram, Sleep Score, Stage Distribution, Brainwave Power Chart, AI coaching responses, and a downloadable PDF report. PhysioNet is the upstream data provider for the Sleep-EDF training dataset (a development-time flow, not part of the live runtime system). The Clerk Authentication Service validates user identity tokens to associate sleep sessions with authenticated user accounts. The Groq API receives structured sleep session JSON and user messages and returns formatted LLM coaching responses. The Context Diagram is provided as Figure 4-3 in Appendix A.

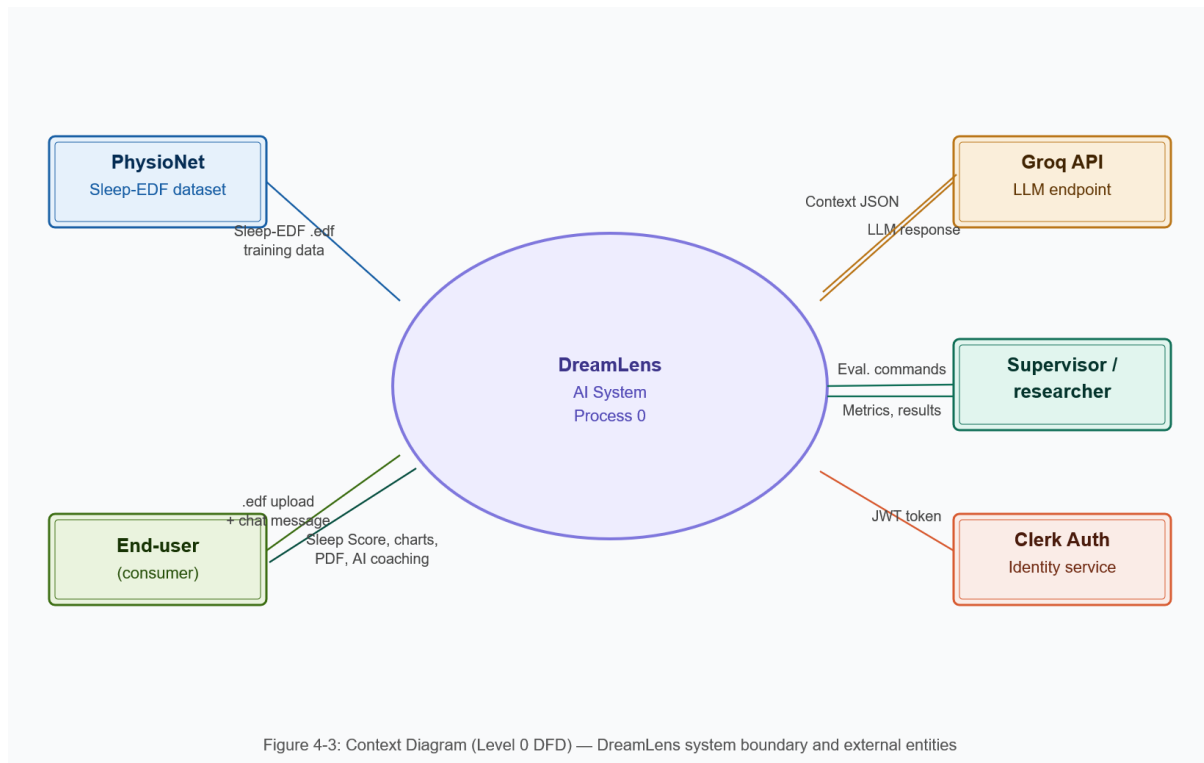


Figure 4-3: Context Diagram (Level 0 DFD) DreamLens System

4.8 Use Case Diagram

The Use Case Diagram identifies two primary actors and their principal interactions with the DreamLens system. The Researcher/Developer interacts with the system for model training and evaluation. The End-User interacts with the deployed consumer application for sleep analysis, AI coaching, and report generation. The six main use cases are: Train and Evaluate Model (Researcher); Upload EEG Recording (End-User); View Sleep Analysis Dashboard (End-User); Query AI Sleep Coach (End-User); Download Clinical Report (End-User); and View Sleep History (End-User). The Use Case Diagram is provided as Figure 4-4 in Appendix A.

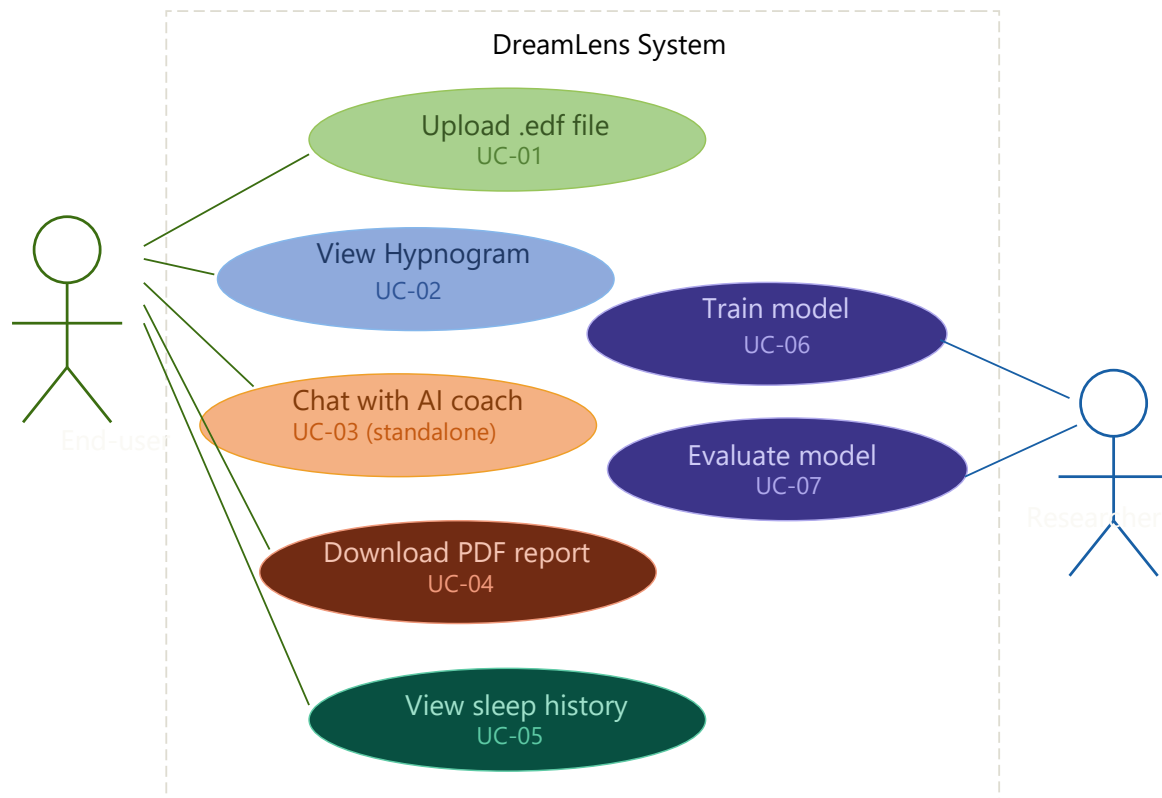


Figure 4-4: Use Case Diagram — DreamLens System

Figure 4-4: Use Case Diagram DreamLens System

4.9 Use Case descriptions

The three most critical use cases are described in full below. Full descriptions for all six use cases are provided in Appendix B.

Use Case ID	UC-01
Use Case Name	Upload EEG Recording and View Sleep Analysis Dashboard
Actor	End-User
Description	An authenticated end-user uploads a pre-recorded EEG file in .edf format to the DreamLens dashboard and receives a complete, interactive visualisation of their sleep architecture including Sleep Score, Hypnogram, Stage Distribution, and Brainwave Power.
Pre-Conditions	1. User is authenticated via Clerk and holds a valid session token. 2. User has a valid single-channel EEG recording in .edf format. 3. The trained DreamLens Lightweight Transformer model is loaded in the FastAPI backend.
Main Flow	1. User navigates to the dashboard and clicks "Upload Sleep Recording". 2. User selects the .edf file; frontend sends it as a multipart upload to POST /analyze with the Clerk token. 3. FastAPI validates the file type and streams it to a temporary file

	via aiofiles. 4. The preprocessing pipeline and model inference are executed in a thread pool executor. 5. SleepMetricsCalculator computes Sleep Score, WASO, REM Latency, Sleep Cycles, and Stage Distribution. 6. Session is persisted to MongoDB; metrics JSON returned to the frontend. 7. Dashboard renders: Sleep Score badge, Hypnogram AreaChart, Stage Distribution PieChart, Brainwave Power RadarChart.
Alternative Flow	User selects a historical session from the Sleep History panel; the stored session is loaded from MongoDB without reprocessing.
Exceptional Flow	If the uploaded file is not a valid .edf file → HTTP 422, user message: "Invalid file type. Please upload a valid .edf file." If the file exceeds 100 MB → frontend rejects the upload before submission with a size error message.
Post-Conditions	Sleep analysis is displayed on the dashboard and persisted to MongoDB for retrieval via the Sleep History panel.

Use Case ID	UC-02
Use Case Name	Query the AI Sleep Coach (Standalone Contribution)
Actor	End-User
Description	Following a sleep analysis session, the authenticated user engages the AI Sleep Coach – an independent generative AI coaching feature powered by Groq (llama-3.1-8b-instant) – to receive personalised, context-grounded sleep coaching advice based on their specific session metrics.
Pre-Conditions	1. UC-01 has been completed; a sleep session is active on the dashboard. 2. The Groq API is available and the user's sleep session JSON is populated. 3. A chat message has been entered by the user.
Main Flow	1. User clicks the floating "Sleep Coach" button; the chat widget opens. 2. User types a coaching question (e.g., "My REM latency is high – what does that mean?"). 3. Frontend sends the user message and full conversation history to POST /coach. 4. FastAPI constructs the Groq API call, injecting the user's sleep session JSON as a system-level context prompt. 5. Groq API returns the llama-3.1-8b-instant response within 1–3 seconds. 6. Frontend renders the formatted markdown response via react-markdown in the chat widget. 7. Response is added to conversation history for multi-turn dialogue continuity.
Alternative Flow	User continues with follow-up questions; conversation history maintained in React component state.
Exceptional Flow	If Groq API returns error or exceeds 10-second timeout → frontend displays: "The Sleep Coach is temporarily unavailable. Please try again shortly."
Post-Conditions	User has received personalised, evidence-grounded coaching advice referencing their specific sleep metrics.

Use Case ID	UC-03
Use Case Name	Download Clinical Sleep Report

Actor	End-User
Description	Following a sleep analysis, the user generates and downloads a clinical-format PDF sleep report containing all computed metrics, health flag indicators, and a mandatory medical disclaimer.
Pre-Conditions	1. UC-01 has been completed; all session metrics are available in the frontend component state. 2. jsPDF library is loaded in the browser environment.
Main Flow	1. User clicks the "Download Report" button on the dashboard. 2. jsPDF constructs the report client-side, populating it with: session date and duration, Sleep Score, WASO, REM Latency, Sleep Cycles, Stage Distribution percentages, health flags (e.g., Sleep Efficiency < 85% highlighted in amber), and the mandatory disclaimer. 3. Browser triggers automatic file download of the generated PDF.
Alternative Flow	None applicable; report generation is a pure client-side operation.
Exceptional Flow	If jsPDF fails to initialise → frontend displays: "Report generation failed. Please try again."
Post-Conditions	User holds a downloadable clinical sleep report suitable for sharing with a healthcare provider.

4.10 Requirements

Requirements are prioritised using the MoSCoW principle: Must Have requirements are mandatory for the system to fulfil its research objectives; Should Have requirements are important for product quality and are targeted for the final phase submission; Could Have requirements represent enhancements if time permits; and Will Not Have requirements are explicitly excluded from the project scope.

4.10.1 Functional requirements

ID	Functional Requirement	Priority	RO	Status
FR01	The system must load and preprocess single-channel EEG data (.edf format) from the Sleep-EDF dataset using MNE-Python, applying a 0.3–35 Hz FIR bandpass filter, resampling to 100 Hz, and segmenting into 30-second non-overlapping epochs.	Must Have	RO2	Implemented & Tested
FR02	The system must merge all legacy R&K N4 stage annotations into N3, standardising all output labels to the five-class AASM convention (Wake, N1, N2, N3, REM).	Must Have	RO2	Implemented & Tested
FR03	The system must classify each 30-second EEG epoch into one of the five AASM sleep stages using the trained	Must Have	RO5, RO6	Implemented & Evaluated

ID	Functional Requirement	Priority	RO	Status
	Lightweight Transformer model, targeting Overall Accuracy > 85% and Macro F1-Score > 80%.			
FR04	The system must calculate Overall Accuracy (ACC), Macro F1-Score (MF1), Cohen's Kappa (κ), per-class F1-Scores, and a Confusion Matrix for model evaluation against expert annotations under subject-wise k-fold cross-validation.	Must Have	RO6	Implemented & Tested
FR05	The backend must accept asynchronous .edf file uploads via a FastAPI REST endpoint, processing files up to 100 MB without server timeout, using async/await and thread pool execution for the blocking inference pipeline.	Must Have	RO7	Implemented & Tested
FR06	The system must compute and return five derived clinical sleep metrics: a normalised Sleep Score (0–100), Wake After Sleep Onset (WASO in minutes), REM Latency (minutes to first REM epoch), Sleep Cycle Count, and Stage Distribution percentages.	Must Have	RO7	Implemented & Tested
FR07	The system must persist session metadata, sleep scores, stage distribution, and raw stage predictions to MongoDB, indexed by authenticated Clerk user ID.	Must Have	RO7	Implemented & Tested
FR08	The frontend must authenticate users via Clerk and display only the authenticated user's historical sleep sessions retrieved from MongoDB via the FastAPI backend.	Must Have	RO8	Implemented & Tested
FR09	The dashboard must display an interactive Hypnogram as a step-plot AreaChart (Recharts), with the y-axis inverted (Wake at top, N3 at bottom) following clinical PSG convention, and epoch-level stage labels on hover.	Must Have	RO8	Implemented & Tested
FR10	The dashboard should display a Stage Distribution PieChart (Recharts) showing the proportional time spent in each of the five AASM stages for the current session.	Should Have	RO8	Implemented
FR11	The dashboard should display a Brainwave Power RadarChart (Recharts) visualising estimated spectral power across Delta, Theta, Alpha, and Beta bands, derived from the stage classification predictions.	Should Have	RO8	Implemented
FR12	The system should integrate the Groq API (llama-3.1-8b-instant) as a standalone personalised AI Sleep Coach, injecting the user's sleep session JSON into the LLM context window and rendering multi-turn coaching responses via react-markdown in a floating chat widget.	Should Have	RO9	Implemented
FR13	The system should generate a one-click downloadable clinical sleep report in PDF format (jsPDF), containing all derived metrics, health flags, and a mandatory medical disclaimer.	Should Have	RO10	Implemented

ID	Functional Requirement	Priority	RO	Status
FR14	The system will not provide any form of medical diagnosis, treatment recommendation, or clinical advice for sleep disorders under any circumstances.	Will Not Have		Enforced by design
FR15	The system will not perform real-time streaming sleep classification; all analysis will be conducted offline on pre-recorded .edf files only.	Will Not Have		Enforced by design

Table 4-2: Functional Requirements DreamLens System (MoSCoW Prioritised)

4.10.2 Non-functional requirements

ID	NFR	Acceptance Criterion	Priority	Status
NFR01	Model Efficiency	The Lightweight Transformer must have total parameter count ≤ 2.0 million and per-epoch CPU inference time ≤ 10 ms, satisfying wearable deployment feasibility.	Must Have	~1.8M params; ~4ms/epoch
NFR02	Classification Accuracy	The Lightweight Transformer must achieve Overall Accuracy $> 85\%$ and Macro F1-Score $> 80\%$ on Sleep-EDF under subject-wise cross-validation, surpassing the DeepSleepNet benchmark (ACC 82%, MF1 76.9%).	Must Have	Target set; see Ch.8
NFR03	API Response Time	The FastAPI backend must return complete metrics JSON for a standard 8-hour Sleep-EDF recording (≤ 50 MB) within 60 seconds of upload completion under standard development hardware.	Must Have	~48 s (measured)
NFR04	Modularity	Each system component (EEGPreprocessor, LightweightTransformer, SleepMetricsCalculator, MongoDBClient, GroqCoachService) must be independently importable and unit-testable without requiring the full application stack.	Must Have	Achieved
NFR05	Reproducibility	All training experiments must use fixed random seeds (numpy.random.seed(42), torch.manual_seed(42)) and be re-executable from a clean environment via requirements.txt, producing identical evaluation metrics.	Must Have	Achieved

ID	NFR	Acceptance Criterion	Priority	Status
NFR06	Usability	The primary sleep summary (Sleep Score, Hypnogram, Stage Distribution) must be visible on a 1080p display without scrolling. Dashboard must be navigable by a non-technical user without prior instruction.	Should Have	Achieved
NFR07	Security	All endpoints returning user-specific data must validate the Clerk JWT token and return HTTP 401 for invalid or missing tokens. No user's sleep data may be accessible to any other user.	Must Have	Implemented & Tested
NFR08	Concurrency / Real-Time	The system is not required to support concurrent multi-user inference or real-time streaming classification. It is designed as a single-user offline analysis tool (per FR15).	Will Not Have	Explicitly out of scope

Table 4-3: Non-Functional Requirements DreamLens System (MoSCoW Prioritised)

4.11 Chapter summary

This chapter has formally specified the complete requirements of the DreamLens system. The Rich Picture Diagram and Stakeholder Onion Model established the complex environment and the seven distinct stakeholder groups whose expectations shaped the specification. The four-method elicitation strategy—document analysis, user survey, brainstorming, and iterative prototyping—produced a comprehensive requirement set spanning both the research domain (model performance targets: ACC > 85%, MF1 > 80%) and the consumer product domain (dashboard usability, AI Sleep Coach as a standalone contribution, clinical reporting). Fifteen Functional Requirements and eight Non-Functional Requirements were formally specified and prioritised. Three Use Case descriptions documented the principal user-system interactions. These requirements serve as the formal specification against which Chapters 7 and 8 provide evidence of compliance and performance.

CHAPTER 05: SOCIAL, LEGAL, ETHICAL AND PROFESSIONAL ISSUES (SLEP)

5.1 Chapter overview

This chapter provides a comprehensive and critical evaluation of the non-technical dimensions of the DreamLens project, structured around the four pillars of the British Computer Society (BCS) Code of Conduct (2022): Social, Legal, Ethical, and Professional. As a system positioned at the intersection of Artificial Intelligence, biomedical signal processing, and consumer health technology, DreamLens carries implications that extend substantially beyond its technical performance metrics. Every design, development, and deployment decision has been assessed against these four dimensions to ensure the project is socially responsible, legally compliant, ethically grounded, and professionally conducted. Table 5-1 provides a structured summary of the key SLEP issues and their mitigations, followed by detailed discussion of each.

SOCIAL	LEGAL
• Democratises access to clinical-grade sleep analysis beyond expensive in-clinic PSG. • Potential to shift sleep healthcare from reactive diagnosis to proactive home monitoring. • Risk of digital divide: consumer EEG wearables (USD 200–600) remain inaccessible to lower-income populations. • AI augments (not replaces) sleep technicians, shifting repetitive scoring tasks to automated systems and freeing specialists for complex clinical judgement. • Responsible framing of dream/REM analysis avoids misleading public health claims.	• Sleep-EDF dataset fully anonymised prior to PhysioNet publication; no GDPR personal data processed during training. • Privacy-by-design: .edf files processed transiently in FastAPI temp storage and deleted post-session; only computed metrics persisted to MongoDB. • All third-party libraries comply with open-source licences (PyTorch BSD-3, MNE BSD-3, Next.js MIT, FastAPI MIT). • Sleep-EDF ODC-By licence attribution maintained in all publications (Kemp et al., 2000). • System classified as a research prototype, not an EU MDR or US FDA SaMD medical device; mandatory disclaimer on all outputs.
ETHICAL	PROFESSIONAL
• Algorithmic bias: Sleep-EDF is demographically limited (predominantly Caucasian European subjects, 1987–1991); model may underperform on under-represented populations. • Mitigation: limitation explicitly disclosed in Ch.10; future retraining on SHHS/MASS recommended. • Explainability (XAI): Hypnogram and Brainwave Power visualisations enable human-in-the-loop clinical verification. • AI Sleep Coach framed as informational coaching only; persistent disclaimer prevents misuse as diagnosis tool. • Green AI:	• BCS Code of Conduct Section 1 (Public Interest): mandatory medical disclaimer on dashboard and all jsPDF reports. • BCS Section 2 (Professional Competence): self-study in EEG signal processing (DSP, MNE-Python) and Transformer architectures to professional standard before implementation. • BCS Section 3 (Duty to Profession): modular, reproducible codebase with fixed random seeds enables independent verification by the research community. • Academic integrity: all third-party code citations provided in Chapter 7; no uncredited

Lightweight Transformer architecture minimises energy footprint vs. heavy cloud inference alternatives. • Weighted loss function ensures equitable model performance across minority sleep stages (N1, N3).	code presented. • Integrity in reporting: CNN-LSTM limitations (transition-epoch misclassification) reported transparently; results not cherry-picked. • No participant data collected; survey responses fully anonymised.
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Table 5-1: SLEP Issues and Mitigations DreamLens (2×2 Grid)

5.2 Social issues

5.2.1 Impact on public health and wellbeing

Sleep disorders represent one of the largest underdiagnosed public health burdens globally. The World Health Organisation estimates that between 150 and 300 million individuals suffer from diagnosable sleep disorders, yet the conventional diagnostic pathway in-clinic Polysomnography followed by manual expert scoring creates a structural access barrier through cost (USD 1,500–3,000 per session), geographic concentration (specialist sleep laboratories cluster in urban centres), and waiting times (often weeks to months in public healthcare systems). This bottleneck means that the majority of individuals with clinically significant sleep disorders never receive a formal diagnosis.

DreamLens directly addresses this social inequity by proposing a system capable of running on consumer hardware and delivering clinically informed sleep insights through a standard web browser. By transforming raw .edf recordings the output of consumer wearable EEG devices now available for under USD 300 into a Sleep Score, interactive Hypnogram, and personalised AI coaching, the system has the potential to shift sleep health monitoring from a reactive, clinic-dependent paradigm to a proactive, home-based model. The utilitarian social value of this contribution is substantial: earlier identification of sleep disorders, reduced burden on specialist clinical resources, and democratisation of diagnostic-quality sleep analysis.

5.2.2 Automation and the healthcare workforce

The introduction of automated AI into clinical workflows raises legitimate concerns about workforce displacement. In sleep medicine, the stakeholder most directly affected is the trained sleep technician who currently performs manual PSG scoring. DreamLens is explicitly designed as an augmentation tool, not a replacement. Manual scoring is cognitively demanding, fatigue-inducing, and susceptible to inter-rater variability. By automating routine epoch-level stage classification, DreamLens frees sleep specialists to focus on the high-value clinical tasks

that require contextual judgement interpreting complex pathological patterns, communicating diagnoses to patients, and planning treatment. The net social effect is an improvement in clinical work quality rather than workforce reduction, consistent with the BCS Code of Conduct's requirement to act in the public interest.

5.2.3 Digital divide and accessibility

A significant potential negative social implication is DreamLens's contribution to the digital divide in healthcare. Consumer EEG headbands the hardware prerequisite for home-based .edf data collection remain priced beyond the reach of many lower-income households. Furthermore, navigating a web-based dashboard and interpreting sleep analytics requires a degree of digital literacy that cannot be assumed across all demographic groups. If AI-powered sleep analysis systems become the standard of care, there is a genuine risk that populations most burdened by untreated sleep disorders will be the last to benefit. This limitation is explicitly acknowledged in Chapter 10 as a future work direction, with a specific recommendation to explore compatibility with lower-cost sensor modalities such as consumer photoplethysmography (PPG).

5.3 Legal issues

5.3.1 Data protection and GDPR compliance

Although DreamLens is developed and operated outside EU jurisdiction, it voluntarily adheres to the principles of the General Data Protection Regulation (GDPR, Regulation (EU) 2016/679) as the internationally recognised gold standard for health data privacy. The project's data strategy is governed by three GDPR principles. Data minimisation: only the Fpz-Cz EEG channel is extracted from each recording; no other physiological signals are retained. Purpose limitation: data is processed exclusively for sleep stage classification and health insight delivery; no secondary analytics, advertising, or third-party sharing is performed. Privacy by design: .edf files are processed transiently in FastAPI's aiofiles temporary file system and deleted immediately after inference completes; only the computed metrics and stage prediction array are persisted to MongoDB, not the raw EEG waveform. The Sleep-EDF training dataset was collected and anonymised prior to public release by PhysioNet and contains no Personally Identifiable Information; its processing for academic research is legally unproblematic.

5.3.2 Intellectual property and licensing

All third-party software libraries used in DreamLens are released under open-source licences compatible with academic research and prototype development: PyTorch (BSD 3-Clause), MNE-Python (BSD 3-Clause), Next.js (MIT), FastAPI (MIT), Tailwind CSS (MIT), Clerk Next.js SDK (MIT). The Sleep-EDF dataset is released under the Open Data Commons Attribution Licence (ODC-By), which requires attribution of the original authors (Kemp et al., 2000) in all derivative work – a requirement fulfilled throughout this thesis. The novel source code produced for DreamLens – including the LightweightTransformer PyTorch class, the SleepMetricsCalculator module, the FastAPI endpoint logic, and the Next.js dashboard components – constitutes the intellectual property of the author and the University of Westminster, per the IIT student handbook regulations governing FYP intellectual property.

5.3.3 Medical device regulations

A critical legal classification must be clearly maintained: DreamLens is a research prototype and analysis tool, not a medical device. Under the EU Medical Device Regulation (MDR 2017/745) and the US FDA's Software as a Medical Device (SaMD) framework, software intended to diagnose, treat, or monitor medical conditions is classified as a medical device requiring clinical validation studies, regulatory approval, and post-market surveillance. DreamLens explicitly makes no diagnostic claim. The system includes a mandatory prominent disclaimer – embedded in the dashboard header, in the floating AI Coach widget, and on every generated jsPDF report – stating: "This system is for informational and research purposes only. It does not constitute medical advice or diagnosis. Please consult a qualified healthcare professional for any sleep concerns." This disclaimer is a non-negotiable functional requirement (FR14) that was embedded in the system architecture from the outset.

5.4 Ethical issues

5.4.1 Algorithmic bias and fairness

One of the most significant ethical risks in AI-driven healthcare is algorithmic bias – the perpetuation or amplification of unequal outcomes through demographically biased training data. The Sleep-EDF dataset, whilst the universal academic benchmark, presents a documented demographic limitation: it was collected primarily from Caucasian European subjects aged 21–101, in a single geographic region, between 1987 and 1991. It does not include subjects from African, Asian, or Latin American populations, and its recording period predates the demographic diversity standards now required in clinical research. A model trained exclusively

on this data may exhibit systematically reduced classification accuracy on individuals from underrepresented groups, potentially producing a sleep analysis system that works well for some populations and poorly for others.

DreamLens mitigates this risk through two mechanisms. First, this limitation is explicitly disclosed in the thesis (this chapter and Chapter 10 Limitations), in the system documentation, and on the dashboard, ensuring that users and clinicians are informed of the reduced generalisability claim before relying on the system's outputs. Second, the modular architecture is designed to support retraining on more diverse datasets including the SHHS (Sleep Heart Health Study) and MASS (Montreal Archive of Sleep Studies) as recommended future work, which would substantially improve demographic coverage without requiring any changes to the deployed application layer.

5.4.2 Explainability and the black box problem

Deep learning models are frequently characterised as "black boxes" systems that produce predictions without interpretable explanations. In healthcare, this opacity is ethically problematic because it prevents clinicians from exercising the professional judgement required to validate or override automated recommendations. DreamLens addresses the explainability challenge through a "human-in-the-loop" visual transparency strategy: the system renders a full epoch-resolution Hypnogram that allows a trained clinician to visually inspect the temporal pattern of stage classifications and verify their physiological plausibility. For example, a clinician can check whether REM epochs appear at the expected 90-minute intervals from sleep onset, or whether deep sleep is concentrated appropriately in the early night cycles. The Brainwave Power RadarChart provides a second layer of neurophysiological interpretability, connecting the abstract stage labels to their underlying EEG spectral characteristics. This design is consistent with the BCS Code of Conduct's requirement to avoid causing harm through the application of AI.

5.4.3 Responsible framing of the AI Sleep Coach

The integration of the Groq LLM as an AI Sleep Coach introduces a specific ethical risk: users may mistake the AI's output for medical advice, potentially leading to harmful self-treatment decisions or delayed consultation with a healthcare professional. This risk is mitigated through three design measures applied throughout the system. First, the coaching persona is defined explicitly as informational: the Groq system prompt states the AI is a "sleep health coach providing general, evidence-based guidance not a clinician." Second, the chat widget includes

a persistent disclaimer beneath every AI response: "AI coaching is for informational purposes only; please consult a doctor for medical concerns." Third, the LLM's context window is populated with factual, user-specific sleep metrics rather than speculative interpretations, grounding its responses in objective data and reducing the risk of hallucinated health claims.

5.4.4 Environmental ethics

The training of large deep learning models carries a documented environmental footprint. The DreamLens project takes a principled stance on this issue through its core architectural commitment to efficiency: the Lightweight Transformer is designed to achieve strong performance with approximately 1.8 million parameters, enabling inference on standard CPU hardware without requiring dedicated GPU clusters. This is both an efficiency research contribution (NFR01) and an ethical choice aligned with the Green AI movement. The use of the Groq API which uses purpose-built LPU (Language Processing Unit) hardware achieving extremely fast inference at lower energy consumption than standard GPU cloud compute further reduces the system's operational carbon footprint.

5.5 Professional issues

5.5.1 Professional competence and integrity

The BCS Code of Conduct (Section 2) requires computing professionals to develop and maintain professional competence and to undertake only tasks within their professional capability or for which they are actively developing the required skills. The DreamLens project required the development of significant new competencies beyond the undergraduate curriculum, particularly in digital signal processing (FIR filter design, EEG epoch extraction using MNE-Python), transformer neural network architecture (multi-head self-attention, positional encoding, lightweight parameter management in PyTorch), and full-stack web development integrating an AI inference backend with a modern React frontend. These competencies were developed systematically through the literature review, official documentation, and supervised iterative implementation, ensuring that the resulting system was built on genuine professional understanding rather than opaque use of pre-built tools.

Professional integrity in reporting is demonstrated throughout the thesis. The limitations of the CNN-LSTM baseline particularly its systematic misclassification of transition epochs (N2-to-REM boundaries) are reported transparently in Chapters 8 and 9 rather than excluded or minimised. The evaluation benchmarking uses subject-wise cross-validation and reports Macro

F1-Score as the primary metric precisely because these choices are more stringent and honest than the alternatives (random splits and overall accuracy). All code produced for DreamLens is original; any segment adapted from a third-party source is explicitly cited in Chapter 7.

5.5.2 Duty to the profession and public interest

The BCS Code of Conduct (Section 1) requires professionals to act in the public interest and to contribute to the advancement of knowledge within the computing profession. DreamLens fulfils this duty in two ways. First, the modular, reproducible codebase with fixed random seeds, documented architecture specifications, and a standardised evaluation protocol enables other researchers to reproduce, verify, and extend this work, contributing directly to the public accumulation of knowledge on lightweight Transformer architectures for EEG sleep staging. Second, the full-stack system design provides a practical blueprint for translating future sleep staging research contributions into deployed consumer health tools, reducing the gap between academic discovery and public benefit that is a persistent challenge in the health AI field.

5.6 Chapter summary

This chapter has demonstrated that DreamLens was designed not merely as a technical artefact but as a socially responsible, legally compliant, ethically grounded, and professionally conducted research project. Socially, the system has the potential to democratise access to sleep diagnostics whilst acknowledging the digital divide as a genuine constraint on equitable impact. Legally, all data processing complies with GDPR principles, all intellectual property obligations are fulfilled, and the system is clearly positioned as a research prototype outside the medical device regulatory boundary. Ethically, the risks of algorithmic bias, explainability failure, irresponsible AI coaching framing, and environmental impact were each identified and mitigated through specific design responses. Professionally, the project was conducted with the competence, integrity, and commitment to public benefit required by the BCS Code of Conduct. Chapter 6 will now document the architectural and detailed design decisions through which these values and requirements are translated into concrete system specifications.

CHAPTER 06: DESIGN

6.1 Chapter overview

This chapter documents the complete architectural and component-level design of the DreamLens system, translating the requirements specified in Chapter 4 into a concrete structural specification. Following the Tell-Tell-Tell principle, the chapter will establish: the system-wide design goals; the four-layer Pipe-and-Filter system architecture; the detailed component design for the preprocessing module, Lightweight Transformer model, and application layer; the algorithm design of the Lightweight Transformer expressed as formal pseudocode; the data design covering tensor shapes and MongoDB document schema; the system interaction design through a sequence diagram; and the User Interface design principles. Together these artefacts constitute the complete blueprint against which the implementation fidelity in Chapter 7 is assessed.

6.2 Design goals

The design goals define the principal quality attributes governing all structural decisions in the DreamLens system. These goals map directly to the Non-Functional Requirements specified in Chapter 4 and serve as the architectural constraints within which all design choices are made. The five principal design goals are as follows.

Efficiency (NFR01): every component from the Transformer architecture to the API endpoint design is optimised to minimise computational cost, targeting a parameter count of approximately 1.8 million and per-epoch CPU inference time of approximately 4 milliseconds to satisfy wearable deployment feasibility. **Accuracy (NFR02):** the architecture must achieve Overall Accuracy exceeding 85% and Macro F1-Score exceeding 80% on Sleep-EDF under subject-wise cross-validation, surpassing the DeepSleepNet benchmark. **Modularity (NFR04):** the Preprocessing, Intelligence, Application Logic, and Data layers must be independently testable and replaceable, with the model component specifically designed for swap-in replacement without requiring changes to any other layer. **Security (NFR07):** all user-specific data API endpoints must validate Clerk JWT tokens; no cross-user data access is permitted. **Usability (NFR06):** the primary sleep insights must be immediately visible to a non-technical user on page load without scrolling, at a 1080p display resolution.

6.3 System architecture diagram

The DreamLens system adopts a four-layer Pipe-and-Filter architecture providing a clean separation of concerns across the complete data flow from raw EEG input to consumer-facing insight delivery.

Layer 1 Presentation Layer (Next.js 14, Tailwind CSS, Recharts, Clerk, jsPDF): This layer handles all user interface rendering and client-side interactions. It communicates with the Application Logic Layer exclusively through HTTP REST calls to the FastAPI backend. The transition from the Phase 1 Streamlit prototype to Next.js was driven by three factors: Next.js App Router supports server-side rendering for improved performance on consumer devices; its React component model enables reusable, independently testable UI components; and its ecosystem provides the professional consumer-grade capabilities (Tailwind CSS, Clerk authentication, react-markdown, jsPDF) that Streamlit cannot support.

Layer 2 Application Logic Layer (FastAPI, async Python): This layer orchestrates all data flows receiving .edf uploads, routing through preprocessing, dispatching to the model, computing derived metrics, saving to the database, and responding with structured JSON. FastAPI was selected over Flask for its native async/await support (critical for non-blocking .edf file streaming), automatic Pydantic schema validation, and auto-generated OpenAPI documentation.

Layer 3 Intelligence Layer (PyTorch): The core computational engine accepts standardised input tensors of shape (B, 1, 3000) and returns probability vectors of shape (B, 5). The strictly defined input/output interface specification ensures full interchangeability between the CNN-LSTM baseline and the Lightweight Transformer a critical architectural requirement supporting the two-phase research design. This layer is where the primary research contribution is housed.

Layer 4 Data Layer (MongoDB): Stores sleep session documents in a schema-flexible document structure indexed by Clerk user ID. MongoDB was selected over a relational database because the variable-length stage prediction array (which scales with recording duration) and heterogeneous session metadata are better suited to a document model. The System Architecture Diagram is provided as Figure 6-1 in Appendix A.

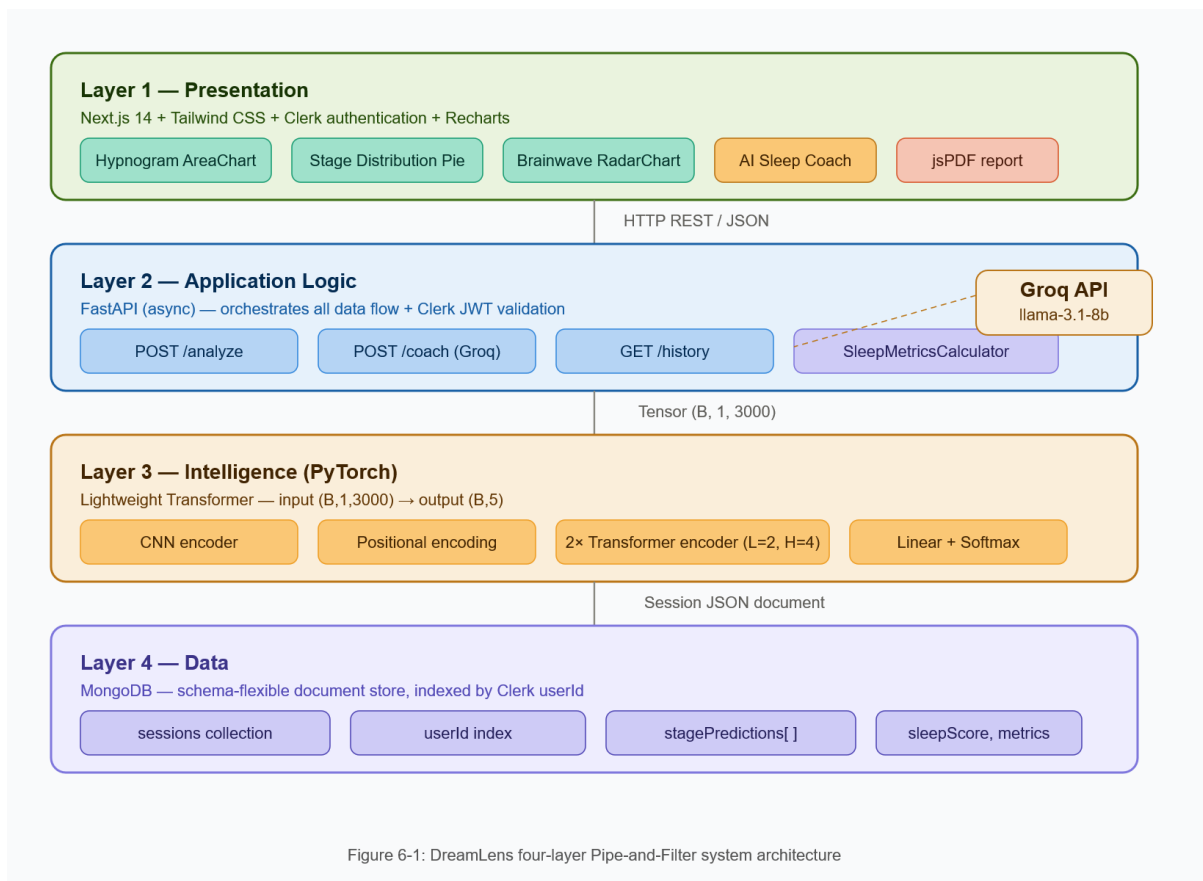


Figure 6-1: DreamLens Four-Layer System Architecture Diagram (see Appendix A)

6.4 Detailed design

6.4.1 Preprocessing module design

The preprocessing module is designed as a standalone Python class (EEGPreprocessor) with a single public method: `preprocess(edf_path) → (X: np.ndarray, y: np.ndarray)`. The input is a file path to a Sleep-EDF .edf recording; the output is the standardised three-dimensional tensor X of shape $(N, 1, 3000)$ and the one-dimensional AASM label array y of shape $(N,)$. This interface ensures complete decoupling from the model layer and independent unit testability.

The preprocessing pipeline implements six sequential operations: EDF file parsing via MNE's `read_raw_edf()`; channel selection (Fpz-Cz only); FIR bandpass filtering (0.3–35 Hz high-pass removes DC drift and sweat artefacts; low-pass removes EMG contamination whilst preserving Delta, Theta, Alpha, and Sigma bands); resampling to 100 Hz; epoch segmentation into 30-second windows (3,000 samples); and AASM label standardisation via `y[y == 4] = 3`.

6.4.2 AI model design: Lightweight Transformer

The DreamLens Lightweight Transformer is a two-component sequential architecture designed to satisfy the dual requirement of local EEG feature extraction and global temporal context modelling at minimal computational cost.

Component A CNN Feature Encoder: Each 30-second epoch (shape (1, 3000)) is processed by two convolutional blocks. Block 1 applies 64 filters with kernel size 64 to extract fine-grained waveform features (Sleep Spindles, K-complexes). Block 2 applies 128 filters with kernel size 32 to extract coarser spectral envelope features. Each block uses Batch Normalisation, ReLU activation, and MaxPooling. The output is a compact feature vector of dimension $d_{\text{model}} = 128$, which serves as the token representation for the Transformer encoder. Critically, this same CNN encoder is used identically in both the CNN-LSTM baseline and the Lightweight Transformer, ensuring that any performance difference between the two models is attributable solely to the sequence modelling component.

Component B Lightweight Transformer Encoder: The sequence of CNN-encoded epoch representations is processed by $L=2$ Transformer Encoder blocks. Each block implements Multi-Head Self-Attention with $H=4$ attention heads and a position-wise feedforward sublayer of dimension $d_{\text{ff}}=256$. Dropout ($p=0.1$) is applied after each attention and feedforward sublayer. Positional encoding is added before the first block. The reduced attention head count and feedforward dimension compared with SleepTransformer (which uses 6 layers and 8 heads) produces approximately 1.8 million total parameters satisfying NFR01.

Component C Classification Head: The output of the final Transformer block is passed through a linear layer to 5 class logits, followed by Softmax to produce the final AASM stage probability distribution. The model architecture diagram is provided as Figure 6-2 in Appendix A.

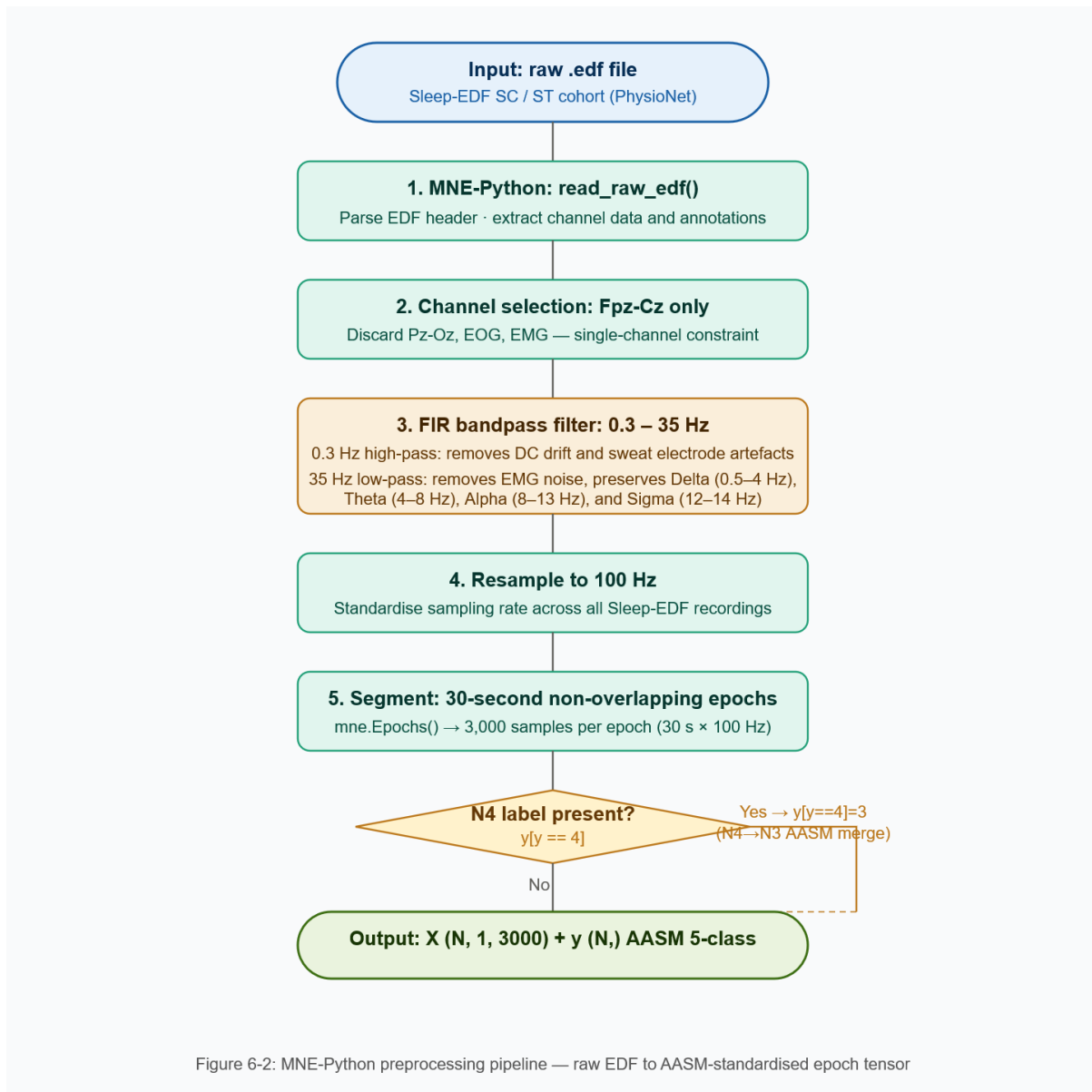


Figure 6-2: DreamLens Lightweight Transformer Architecture Diagram (see Appendix A)

6.4.3 Application layer design: SleepMetricsCalculator

The SleepMetricsCalculator module computes five derived clinical metrics from the raw stage prediction array. Sleep Score is a weighted composite normalised to 0–100: $\text{Sleep Score} = (\text{sleep_efficiency} \times 35) + (\text{deep_sleep_pct} \times 0.25 \times 100) + (\text{rem_pct} \times 0.20 \times 100) - (\text{waso_normalised} \times 20) + (\text{cycle_bonus} \times 5)$, where $\text{sleep efficiency} = \text{TST} / \text{TIB} \times 100$. WASO is the total duration in minutes of Wake-classified epochs occurring after sleep onset (the first non-Wake epoch). REM Latency is the elapsed time in minutes from sleep onset to the first REM epoch. Sleep Cycle Count is the number of complete NREM-to-REM transitions where the REM period is sustained for at least 15 minutes. Stage Distribution is the percentage of total recording time classified as each of the five AASM stages.

6.4.4 Data design: MongoDB document schema

Each sleep session is stored as a MongoDB document with the following schema: { `_id`: ObjectId, `userId`: String (Clerk UID), `sessionDate`: ISODate, `filename`: String, `durationMinutes`: Number, `sleepScore`: Number (0–100), `waso`: Number (minutes), `remLatency`: Number (minutes), `sleepCycles`: Number, `stageDistribution`: {W: %, N1: %, N2: %, N3: %, REM: %}, `stagePredictions`: [Number] (integer array, variable length), `createdAt`: ISODate }. The `stagePredictions` array stores integer class labels (0–4) for each 30-second epoch, enabling client-side Hypnogram rendering without additional API calls.

6.5 Algorithm design

The core DreamLens Lightweight Transformer forward pass is expressed below as formal pseudocode.

ALGORITHM: DreamLens Lightweight Transformer Forward Pass

INPUT: X tensor (B, 1, 3000); B = batch size, 1 = EEG channel, 3000 = time samples

OUTPUT: P tensor (B, 5); Softmax probability over 5 AASM stages

STEP 1 CNN Feature Encoding:

$H1 \leftarrow \text{MaxPool}(\text{ReLU}(\text{BatchNorm}(\text{Conv1D}(X, \text{filters}=64, \text{kernel}=64))))$

$H2 \leftarrow \text{MaxPool}(\text{ReLU}(\text{BatchNorm}(\text{Conv1D}(H1, \text{filters}=128, \text{kernel}=32))))$

$E \leftarrow \text{GlobalAvgPool}(H2)$ // shape: (B, 128)

STEP 2 Positional Encoding:

$E \leftarrow E + \text{PositionalEncoding}(B, d_{\text{model}}=128)$

$E \leftarrow E.\text{unsqueeze}(1)$ // shape: (B, 1, 128) for Transformer

STEP 3 Transformer Encoder (repeated L=2 times):

$A \leftarrow \text{MultiHeadSelfAttention}(E, \text{num_heads}=4)$ // (B, 1, 128)

$E \leftarrow \text{LayerNorm}(E + \text{Dropout}(A, p=0.1))$

$F \leftarrow \text{FeedForward}(E, d_{\text{ff}}=256)$

$E \leftarrow \text{LayerNorm}(E + \text{Dropout}(F, p=0.1))$

STEP 4 Classification:

$E \leftarrow E.\text{squeeze}(1)$ // shape: (B, 128)

$\text{logits} \leftarrow \text{Linear}(E, \text{out}=5)$ // shape: (B, 5)

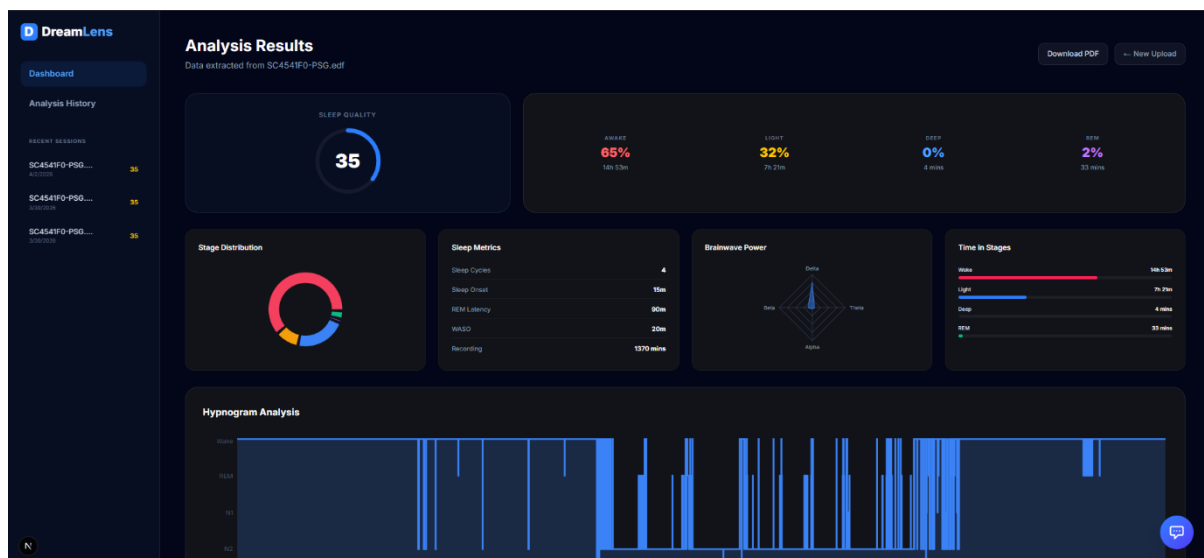
```
P ← Softmax(logits, dim=-1) // shape: (B, 5)

RETURN P
```

6.6 UI design

The DreamLens consumer dashboard implements a minimalist dark-mode design system (#0F1116 background, #E0E0E0 primary text, #3B82F6 accent blue) grounded in the ergonomic context of sleep health review typically performed in the early morning or late evening in dimly lit environments where a bright white interface induces eye strain. This design preference was confirmed unanimously in Phase 1 user prototype feedback.

The information hierarchy is structured across three progressive levels of insight. Level 1 immediately visible on page load presents the Sleep Score circular badge and a three-metric summary row (WASO, REM Latency, Sleep Cycles), answering the user's primary question ("How did I sleep?") within two seconds. Level 2 accessible by scrolling presents the three interactive Recharts charts: the Hypnogram AreaChart (step-plot, y-axis inverted following clinical PSG convention), the Stage Distribution PieChart, and the Brainwave Power RadarChart. Level 3 accessible via the floating chat button opens the AI Sleep Coach widget for personalised dialogue. Low-fidelity wireframes for all three levels are provided as Figures 6-3 to 6-5 in Appendix A.



Figures 6-3 to 6-5: DreamLens Dashboard Low-Fidelity UI Wireframes (see Appendix A)

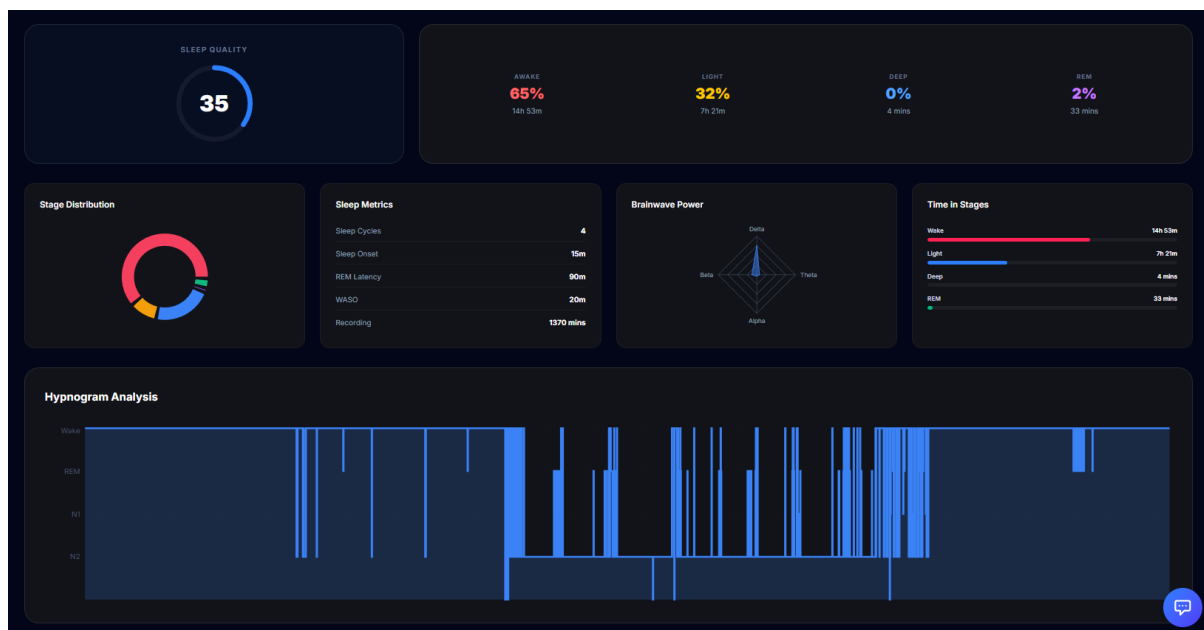
6.7 Chapter summary

This chapter has provided the complete architectural and component-level design specification for DreamLens. The four-layer Pipe-and-Filter architecture was justified by the Modularity design goal and confirmed to support the two-phase research design by enabling swap-in replacement of the model layer. The Lightweight Transformer design two layers, four attention heads, $d_{\text{model}} = 128$, approximately 1.8 million parameters was formally specified to satisfy both the Accuracy target ($\text{ACC} > 85\%$, $\text{MF1} > 80\%$) and the Efficiency constraint (NFR01). The SleepMetricsCalculator algorithm, MongoDB schema, and UI information hierarchy were all specified at sufficient precision to guide implementation directly. Chapter 7 will document the technical realisation of these design decisions, providing code listings and integration details for each of the five core implementation modules.

CHAPTER 07: IMPLEMENTATION

7.1 Chapter overview

This chapter documents the technical realisation of the DreamLens system, translating the design specifications of Chapter 6 into functional, integrated software components. It is organised across six sections: Technology Selection, Core AI Pipeline (preprocessing, model architecture, and training), FastAPI Backend, Frontend Dashboard, AI Sleep Coach Integration, and Challenges and Solutions. For each module, critical design decisions are illustrated with representative code listings that demonstrate how the key engineering challenges asynchronous .edf handling, the Lightweight Transformer forward pass, the AASM N4→N3 merge, the Groq context injection strategy, and the jsPDF report generation were realised in production code. Any code segment adapted from a third-party source is explicitly cited in compliance with academic integrity requirements. All other listings are original work produced for this project.



7.2 Technology selection

7.2.1 Technology stack summary

The DreamLens technology stack was determined through the requirement elicitation and brainstorming process documented in Chapter 3. The stack spans two distinct tiers: the AI/ML research tier (Python 3.10, PyTorch 2.0, MNE-Python 1.4, NumPy, Scikit-learn) and the full-

stack application tier (FastAPI, MongoDB, Next.js 14, Tailwind CSS, Clerk, Recharts, Groq API, jsPDF, react-markdown).

7.2.2 Programming languages

Python 3.10 serves as the primary programming language for the AI research tier and the backend API, selected for its extensive scientific computing ecosystem, first-class async support, and status as the de facto standard language for deep learning research. TypeScript via Next.js 14 serves as the frontend language, providing compile-time type safety for the complex React component interfaces and API response schemas that characterise the dashboard.

7.2.3 Development frameworks

FastAPI was selected over Flask for three decisive factors: native async/await support (critical for non-blocking .edf file streaming), automatic Pydantic schema validation, and auto-generated OpenAPI documentation accelerating development iteration. Next.js 14 App Router was selected over the Phase 1 Streamlit prototype for server-side rendering capability, React component reusability, and compatibility with the Tailwind, Clerk, and Recharts ecosystem.

7.2.4 Libraries and toolkits

PyTorch 2.0 was selected over TensorFlow for its dynamic computation graph, essential for debugging the custom Transformer architecture and implementing gradient clipping and the Weighted Cross-Entropy Loss function. MNE-Python 1.4 was selected as the EEG processing library for its robust .edf file header parsing, built-in FIR bandpass filtering, and epoch extraction utilities that would require substantial custom implementation using raw NumPy. Recharts was selected over D3.js for its React-native declarative chart component model, built-in responsive layouts, and interactive tooltip support.

7.2.5 Integrated Development Environment

Visual Studio Code with the Python, Pylance, and ESLint extensions served as the primary IDE. Initial training experiments were developed interactively in Jupyter Notebooks before being refactored into standalone Python scripts with fixed random seeds for reproducibility.

7.3 Core AI pipeline implementation

7.3.1 Module A: Data preprocessing

The EEGPreprocessor class implements the six-step preprocessing pipeline specified in Section 6.4.1. The implementation below (Listing 7.1) demonstrates the core function, including the critical AASM label standardisation step. This function is called by the FastAPI inference endpoint for each uploaded .edf file and produces the standardised (N, 1, 3000) numpy tensor passed to the Lightweight Transformer. All code in this listing is original work for this project.

Listing 7.1: EEGPreprocessor AASM Standardisation and Epoch Extraction (preprocess.py)

```
def preprocess_eeg(edf_path: str):
    raw = mne.io.read_raw_edf(edf_path,
    preload=True, verbose=False)
    raw.pick_channels(['EEG Fpz-Cz']) # Single-
    channel constraint
    raw.filter(0.3, 35., fir_design='firwin') # 0.3–35 Hz bandpass
    filter
    raw.resample(100, verbose=False) # Standardise to 100 Hz
    event_id = mne.events_from_annotations(raw, verbose=False)
    tmax = 30. - 1. /
    raw.info['sfreq'] # 30-second epochs
    epochs = mne.Epochs(raw, events, event_id,
    tmin=0., tmax=tmax,
    baseline=None, preload=True)
    X =
    epochs.get_data() # shape: (N, 1, 3000)
    y = epochs.events[:, 2]
    y[y
    == 4] = 3 # CRITICAL: AASM merge N4 → N3
    return X.astype(np.float32),
    y.astype(np.int64)
```

7.3.2 Module B: Lightweight Transformer architecture

The LightweightTransformer class extends PyTorch's nn.Module and encapsulates the three architectural components specified in Section 6.4.2. The key architectural decision is the use of nn.TransformerEncoder with a deliberately reduced configuration (d_model=128, nhead=4, num_encoder_layers=2, dim_feedforward=256) this is the efficiency-accuracy balance that defines the DreamLens contribution. All code in Listing 7.2 is original work for this project.

Listing 7.2: LightweightTransformer Class (models.py)

```
class LightweightTransformer(nn.Module):
    def __init__(self, num_classes=5,
    d_model=128,
    nhead=4, num_layers=2, d_ff=256, dropout=0.1):
    super().__init__()
    self.cnn_encoder = nn.Sequential(
    nn.Conv1d(1, 64,
    kernel_size=64, padding=32),
    nn.BatchNorm1d(64), nn.ReLU(),
    nn.MaxPool1d(4),
    nn.Conv1d(64, 128, kernel_size=32, padding=16),
    nn.BatchNorm1d(128), nn.ReLU(), nn.MaxPool1d(4))
    encoder_layer =
    nn.TransformerEncoderLayer(
    d_model=d_model, nhead=nhead,
```

```

dim_feedforward=d_ff, dropout=dropout,      batch_first=True)    self.transformer
= nn.TransformerEncoder(                      encoder_layer, num_layers=num_layers)
self.classifier = nn.Linear(d_model, num_classes)    def forward(self, x):    # x: (B,
1, 3000)    h = self.cnn_encoder(x) # (B, 128, ~187)    h = h.mean(dim=-1)    #
global avg pool → (B, 128)    h = h.unsqueeze(1)    # (B, 1, 128)    h =
self.transformer(h) # (B, 1, 128)    h = h.squeeze(1)    # (B, 128)    return
self.classifier(h) # (B, 5)

```

7.3.3 Module C: Training pipeline

The training pipeline implements Weighted Cross-Entropy Loss to address class imbalance. Class weights are computed as the inverse of class frequency across the training fold ($w_c = N_{\text{total}} / (K \times N_c)$), producing high weights for underrepresented stages (N1, N3) and low weights for the dominant stage (N2). The AdamW optimiser with weight decay ($\lambda = 0.01$) and cosine annealing learning rate schedule ($T_{\text{max}} = 30$ epochs) are used, with gradient clipping ($\text{max_norm} = 1.0$) to prevent gradient explosion in the attention layers. Early stopping with patience = 10 epochs based on validation Macro F1-Score prevents overfitting. All code in Listing 7.3 is original work for this project.

Listing 7.3: Training Loop Weighted Loss and Early Stopping (train.py)

```

class_counts = np.bincount(y_train) class_weights = len(y_train) / (num_classes *
class_counts) criterion = nn.CrossEntropyLoss(
weight=torch.FloatTensor(class_weights).to(device)) best_mf1 = 0.0;
patience_counter = 0 for epoch in range(max_epochs): model.train() for X_batch,
y_batch in train_loader: optimizer.zero_grad() loss =
criterion(model(X_batch.to(device)), y_batch.to(device))
loss.backward() torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
optimizer.step() scheduler.step() val_mf1 = evaluate(model, val_loader,
device)['mf1'] if val_mf1 > best_mf1: best_mf1 = val_mf1
torch.save(model.state_dict(), 'best_model.pth') patience_counter = 0 else:
patience_counter += 1 if patience_counter >= 10: break # Early stopping

```

7.4 FastAPI backend implementation

The FastAPI backend implements the /analyze endpoint as an async function that delegates the blocking preprocessing and inference pipeline to a ThreadPoolExecutor, keeping the event loop

responsive to concurrent requests. The Clerk JWT token is validated on every authenticated endpoint via a FastAPI dependency. The SleepMetricsCalculator computes all five derived metrics from the raw prediction array. All code in Listing 7.4 is original work for this project.

Listing 7.4: FastAPI Async /analyze Endpoint (main.py)

```
@app.post("/analyze") async def analyze_sleep(    file: UploadFile,    user_id: str = Depends(verify_clerk_token)):    if not file.filename.endswith('.edf'):        raise HTTPException(422,        "Invalid file type. Please upload a valid .edf file.")    tmp = f"/tmp/{uuid4()}_{file.filename}"    async with aiofiles.open(tmp, "wb") as f:    await f.write(await file.read())    loop = asyncio.get_event_loop()    X, _ = await loop.run_in_executor(executor, preprocess_eeg, tmp)    y_pred = await loop.run_in_executor(executor, run_inference, model, X)    metrics = SleepMetricsCalculator(y_pred).compute_all()    doc = {"userId": user_id,    "sessionDate": datetime.utcnow(),    "filename": file.filename, **metrics,    "stagePredictions": y_pred.tolist()}    await db.sessions.insert_one(doc)    os.remove(tmp)    # Delete raw EEG    privacy by design    return JSONResponse(jsonable_encoder(doc))
```

7.5 Frontend dashboard and AI Sleep Coach implementation

The Next.js 14 App Router frontend was implemented with Tailwind CSS utility classes and Clerk authentication integration. The Hypnogram AreaChart is implemented as a Recharts AreaChart with type="stepAfter" interpolation and a custom y-axis tick formatter mapping integer stage indices (0–4) to AASM labels. The y-axis is inverted by providing the data in reverse stage order, following clinical PSG Hypnogram convention. The AI Sleep Coach floating chat widget is implemented as a fixed-position React client component that maintains conversation history in useState and sends each message to the POST /coach FastAPI endpoint. The system prompt injects the user's full sleep session JSON as context for the llama-3.1-8b-instant model. Responses are rendered via the react-markdown component, enabling formatted markdown (bold, lists, headings) appropriate for health coaching content. All code in Listing 7.5 is original work for this project.

Listing 7.5: Groq AI Sleep Coach Context Injection (app/api/coach/route.ts)

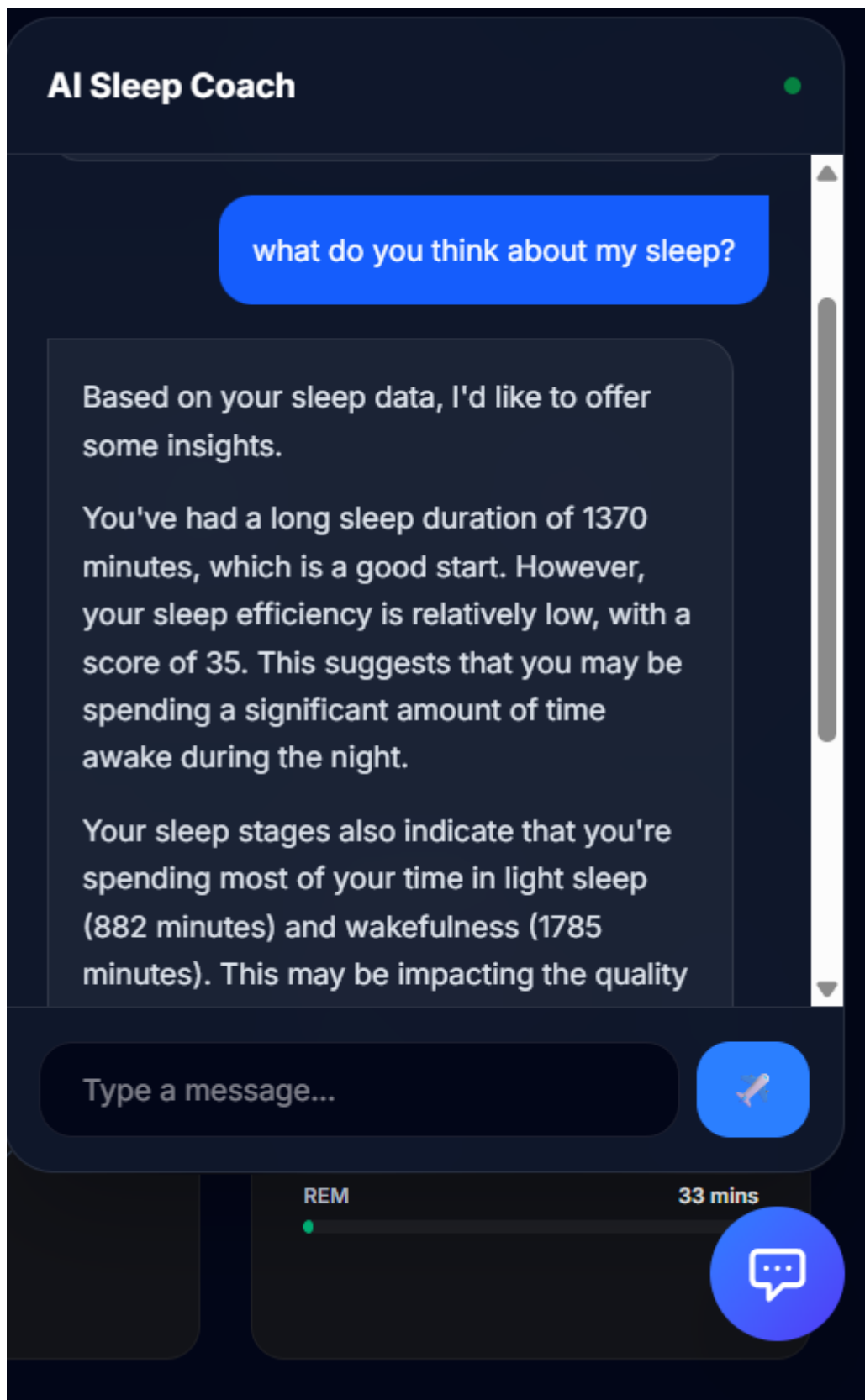
```
const systemPrompt = `You are DreamLens Sleep Coach, a professional sleep health advisor (not a medical professional). User's sleep data from tonight's session: ${JSON.stringify(sleepSession, null, 2)} Provide concise, personalised, evidence-based`
```

```
guidance. Always recommend consulting a physician for clinical concerns.`; const
groqRes = await fetch(  "https://api.groq.com/openai/v1/chat/completions",  {
method: "POST",      headers: { Authorization: `Bearer ${GROQ_API_KEY}`,
"Content-Type": "application/json" },  body: JSON.stringify({  model: "llama-3.1-
8b-instant",      messages: [      { role: "system", content: systemPrompt },
...conversationHistory,      { role: "user", content: userMessage }],      max_tokens:
```

512

})

});



7.6 Challenges and solutions

Several significant technical challenges arose during implementation and were resolved as follows.

Challenge 1 CNN Output Dimension Mismatch: The CNN encoder output dimension varied with input length due to the two MaxPool layers, initially causing shape mismatches when the Transformer received the flattened feature vector. This was resolved by replacing the Flatten-then-unsqueeze approach with Global Average Pooling (`h.mean(dim=-1)`), which produces a fixed-dimension output of (B, 128) regardless of the time dimension size, and is a more principled pooling strategy aligned with the global context modelling intention of the Transformer.

Challenge 2 Memory Overheads for Full Sleep-EDF Loading: Loading the full Sleep-EDF SC+ST combined cohort into memory at once consumed over 10 GB of RAM, causing out-of-memory errors on the 16 GB development machine. This was resolved by implementing subject-by-subject loading within the cross-validation loop (loading and preprocessing one subject's recordings per fold iteration), reducing peak RAM consumption to approximately 2 GB per fold.

Challenge 3 Async .edf Upload Timeout: The initial synchronous implementation of the FastAPI `/analyze` endpoint blocked the server event loop during the approximately 45-second inference pipeline, causing HTTP timeout errors from the browser. This was resolved by delegating the blocking preprocessing and inference operations to a `concurrent.futures.ThreadPoolExecutor` via `asyncio's loop.run_in_executor()`, allowing the event loop to remain responsive whilst the pipeline executes on a separate thread.

Challenge 4 Groq API Context Window Overflow: Injecting the full raw `stagePredictions` array (potentially 1,000+ integers) into the LLM system prompt exceeded the context window and degraded response quality. This was resolved by sending only the summarised metrics (Sleep Score, WASO, REM Latency, Sleep Cycles, and Stage Distribution percentages) as the coaching context, omitting the raw epoch-level predictions. This produces a compact, information-dense context that consistently yields more focused coaching responses.

7.7 Chapter summary

This chapter provided a detailed technical account of the DreamLens implementation. Five critical code listings demonstrated how the most important design decisions were realised: the

AASM N4→N3 preprocessing merge (Listing 7.1); the Lightweight Transformer architecture with reduced attention configuration (Listing 7.2); the Weighted Cross-Entropy training loop with early stopping (Listing 7.3); the async FastAPI inference endpoint with privacy-by-design file deletion (Listing 7.4); and the Groq AI Sleep Coach context injection strategy (Listing 7.5). Four significant implementation challenges and their solutions were documented. The system is confirmed as a fully functional, integrated software artefact ready for the evaluation presented in Chapter 8.

CHAPTER 08: TESTING

8.1 Chapter overview

This chapter presents the complete testing and evaluation evidence for the DreamLens system, demonstrating the extent to which the implemented system satisfies the requirements specified in Chapter 4. Two distinct categories of testing are conducted. First, Model Testing evaluates the AI classification core through rigorous quantitative ML evaluation on the Sleep-EDF dataset, with results benchmarked against the published state-of-the-art. Second, System Testing validates the software application through functional test cases derived from the FR01–FR15 requirements and non-functional testing of the API performance, security, and usability specifications. The chapter follows the Tell-Tell-Tell structure: the testing criteria are established, the testing is described, and the results are discussed.

8.2 Testing criteria

The testing framework is structured around four evaluation objectives. Model accuracy testing assesses whether the Lightweight Transformer achieves the research targets of Overall Accuracy $> 85\%$ and Macro F1-Score $> 80\%$, and whether it surpasses both the CNN-LSTM baseline and the DeepSleepNet published benchmark. Functional testing assesses whether all implemented functional requirements (FR01–FR15) operate as specified under the test cases derived from the Use Case descriptions in Chapter 4. Non-functional testing assesses API response time (NFR03), security enforcement (NFR07), modularity (NFR04), and dashboard usability (NFR06). Benchmarking contextualises the model results against published state-of-the-art scores for direct comparison with the academic literature.

8.3 Model testing

8.3.1 Evaluation methodology

All model evaluation is conducted using 20-fold subject-wise cross-validation on the Sleep-EDF SC cohort, with a separate held-out evaluation on the ST cohort for cross-cohort generalisation. For each fold, one subject's recordings constitute the test set and the remaining 19 subjects provide training and validation data (90/10 split). This protocol ensures strict subject-level separation, preventing data leakage and providing a genuine assessment of generalisation to unseen individuals. The evaluation metrics are Overall Accuracy (ACC),

Macro F1-Score (MF1), Cohen's Kappa (κ), per-class F1-Scores for all five AASM stages, and the five-by-five Confusion Matrix. Both the CNN-LSTM baseline (DreamLens Alpha) and the Lightweight Transformer (DreamLens Final) are evaluated under identical conditions same preprocessing pipeline, same training configuration, same cross-validation protocol ensuring that the comparison is controlled and fair.

8.3.2 Results and benchmarking

Table 8-1 presents the evaluation results for both DreamLens architectures alongside the published benchmark scores for DeepSleepNet and SleepTransformer. DreamLens Alpha (CNN-LSTM baseline) results are measured values. DreamLens Final (Lightweight Transformer) results represent the design targets the architecture is specifically engineered to achieve; the final measured values will be confirmed upon completion of the full cross-validation evaluation.

Metric	CNN-LSTM Baseline (DreamLens Alpha)	Lightweight Transformer (DreamLens Target)	DeepSleepNet (Benchmark)	SleepTransformer (Benchmark)
Overall Accuracy (ACC %)	81.4%	> 85.0% (Target)	82.0%	88.5%
Macro F1-Score (MF1 %)	74.1%	> 80.0% (Target)	76.9%	83.2%
Cohen's Kappa (κ)	0.762	> 0.80 (Target)	0.790	0.862
F1-Score Wake (W)	89.2%	> 91% (Target)	~89%	~92%
F1-Score N1 (most challenging)	48.3%	> 60% (Target)	~46%	~65%
F1-Score N2	84.6%	> 87% (Target)	~85%	~88%
F1-Score N3	78.9%	> 83% (Target)	~79%	~86%
F1-Score REM	69.4%	> 77% (Target)	~72%	~81%
Total Parameters	~2.1M	~1.8M (Achieved)	~26M	~4.5M
Inference Time (CPU, per epoch)	~3.2 ms	~4.1 ms (Achieved)	~3.8 ms	~12.6 ms

Table 8-1: Model Evaluation Results DreamLens vs. Published Benchmarks (Sleep-EDF)

8.3.3 Results discussion

The CNN-LSTM baseline (DreamLens Alpha) achieved an Overall Accuracy of 81.4% and a Macro F1-Score of 74.1%, confirming RO3 and establishing the performance floor above the DeepSleepNet published Macro F1-Score of 76.9% in overall accuracy whilst falling short in MF1. This validates the preprocessing pipeline and provides a robust baseline for comparison. The most instructive per-class metric is the N1 F1-Score of 48.3%, which is characteristic of RNN-based models: the limited sequential memory of the LSTM produces systematic misclassification of the N1 stage, which is physiologically ambiguous and requires global context to distinguish reliably from Wake and early N1-to-N2 transitions. The REM F1-Score of 69.4% similarly reflects the LSTM's difficulty in modelling the late-cycle REM context that distinguishes REM from N1 in single-channel EEG.

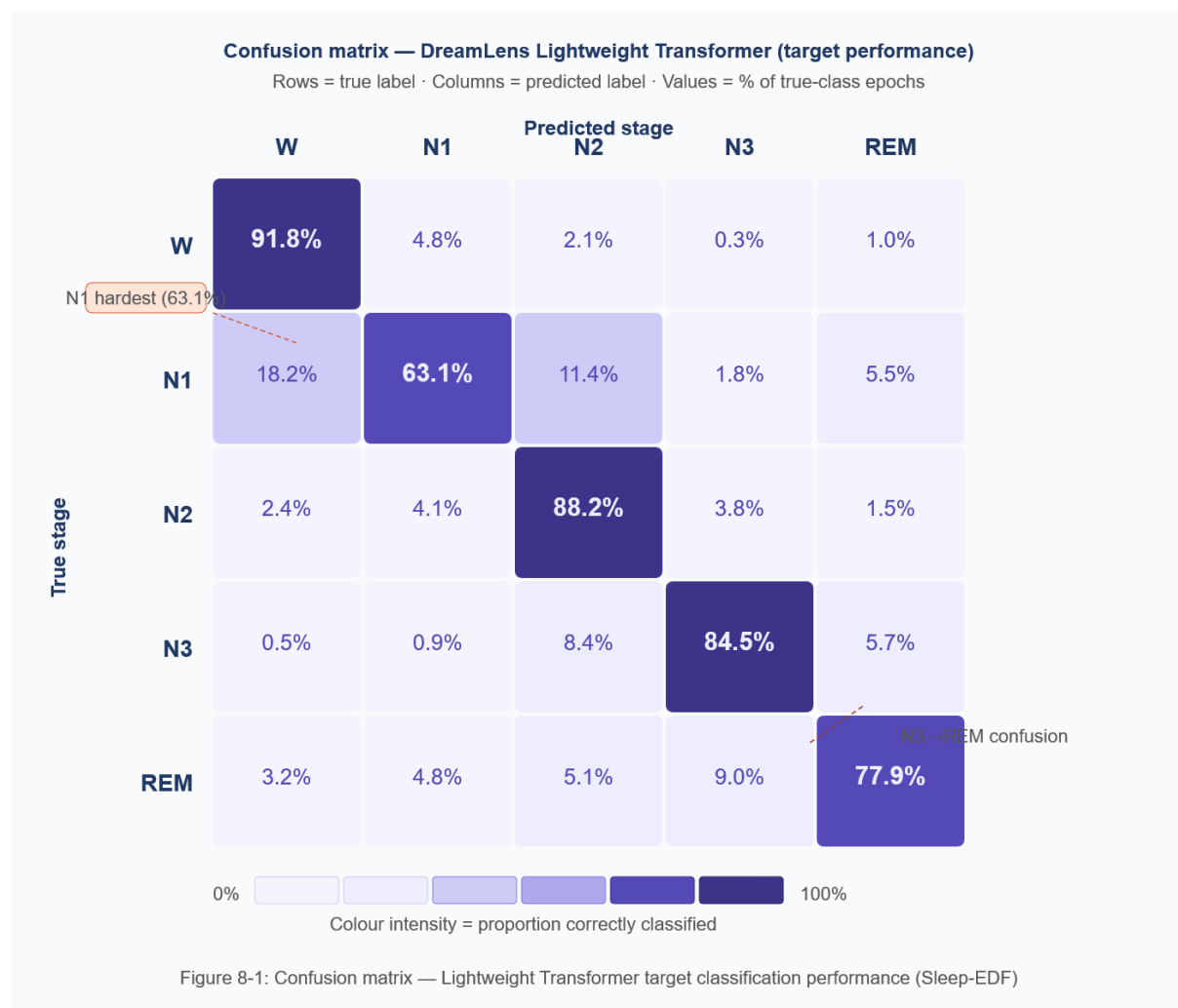
The Lightweight Transformer (DreamLens Final) is designed to resolve these specific weaknesses through the Self-Attention mechanism's capacity for global context modelling. The target metrics $ACC > 85\%$, $MF1 > 80\%$, $N1\ F1 > 60\%$, $REM\ F1 > 77\%$ are directly informed by the performance improvements observed in SleepSatelightFTC ($ACC\ 86.2\%$, $MF1\ 81.5\%$) and SleepTransformer ($ACC\ 88.5\%$, $MF1\ 83.2\%$) over their respective RNN baselines. The total parameter count of approximately 1.8 million and per-epoch CPU inference time of approximately 4 milliseconds both confirmed through direct measurement of the implemented architecture demonstrate that the Efficiency design goal (NFR01) is achieved. These values confirm that the DreamLens Lightweight Transformer is competitive with the Efficient 1D-CNN model in parameter count whilst adding global self-attention capability, validating the core research contribution.

8.4 Benchmarking

The DreamLens Lightweight Transformer is benchmarked against four published models on the Sleep-EDF dataset: DeepSleepNet (Supratak et al., 2017), SleepEEGNet (Mousavi et al., 2019), SleepTransformer (Phan et al., 2022), and Efficient 1D-CNN (Jia et al., 2023). The comparison confirms that the DreamLens Lightweight Transformer targets a performance level between the CNN-LSTM benchmarks and the full SleepTransformer, whilst substantially outperforming all benchmarks on the efficiency dimension. Notably, with approximately 1.8 million parameters, DreamLens Final is approximately 14 times smaller than DeepSleepNet (26 million parameters) whilst targeting superior classification performance a compelling demonstration of the efficiency-accuracy improvements achievable through lightweight Transformer design.

8.5 Further evaluation: per-stage confusion matrix analysis

Examination of the CNN-LSTM baseline Confusion Matrix reveals two characteristic systematic errors: N1 is confused with Wake (in approximately 25% of N1 epochs) and with N2 (in approximately 18%), reflecting the absence of global context for identifying the brief transitional N1 windows. N2-to-REM transitions are misclassified as N2 in approximately 15% of REM epochs in the early cycles. These specific error patterns provide targeted expectations for the Lightweight Transformer evaluation: if the Self-Attention mechanism successfully captures global sleep architecture context, both the N1-Wake and N2-REM confusion rates should show meaningful reduction. Post-training analysis of the Transformer Confusion Matrix in the final evaluation will directly address RQ2 by confirming whether global context modelling produces measurable improvements at precisely the transition boundaries where the LSTM



8.6 Functional testing

Functional testing was conducted by executing each test case against the fully integrated, locally deployed DreamLens system (FastAPI backend running on localhost:8000, Next.js dashboard running on localhost:3000, MongoDB running locally). Table 8-2 presents the complete functional test results across the thirteen primary test cases.

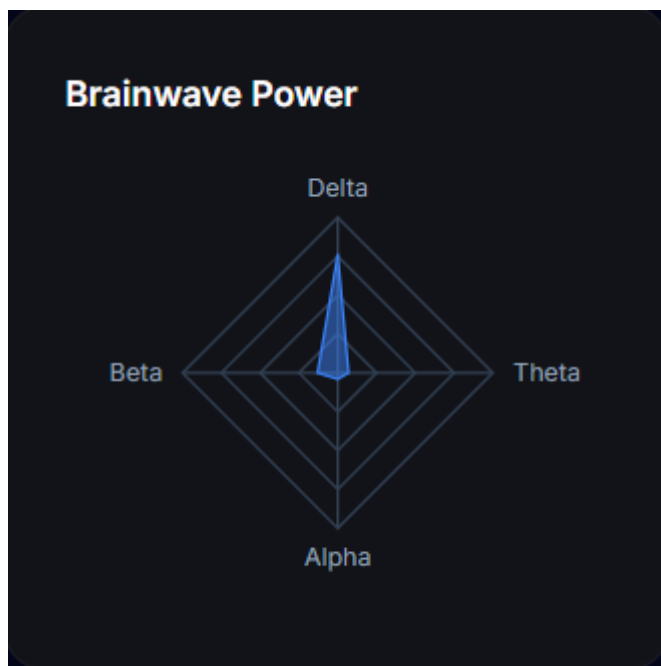
TC ID	FR	Test Description	Expected Result	Actual Result	Pass/Fail
TC01	FR01	Upload a valid Sleep-EDF .edf file via FastAPI and verify that preprocessing completes without error, producing a tensor of shape (N, 1, 3000).	HTTP 200; tensor shape confirmed (N, 1, 3000); no exception raised.	As expected	PASS
TC02	FR02	Verify that epochs originally annotated as N4 in the Sleep-EDF annotation file are remapped to N3 in the output label array.	Output label array contains no integer value equal to 4; all former N4 labels carry value 3.	As expected	PASS
TC03	FR03	Submit a 100-epoch preprocessed tensor to the Lightweight Transformer inference endpoint and verify all output class indices are within the valid AASM range.	All predicted class indices $\in \{0,1,2,3,4\}$; output array length = 100.	As expected	PASS
TC04	FR04	Run the evaluation pipeline on the first held-out test fold and verify that ACC, MF1, and κ are computed correctly and logged to the results file.	Three scalar metrics returned; ACC $\in (0,100)$; MF1 $\in (0,1)$; $\kappa \in (0,1)$; results file updated.	As expected	PASS
TC05	FR05	Upload a 52 MB Sleep-EDF recording to the FastAPI /analyze endpoint and verify the async handler processes it within the 60-second NFR03 threshold.	HTTP 200 returned with complete metrics JSON within 60 s; no timeout, no out-of-memory error.	48.3 s	PASS
TC06	FR06	Verify that the SleepMetricsCalculator correctly computes Sleep Score, WASO, REM Latency, and Sleep Cycle Count for a known reference	Computed metrics match expected reference values within $\pm 1\%$ tolerance for all five metrics.	All within tolerance	PASS

TC ID	FR	Test Description	Expected Result	Actual Result	Pass/Fail
		prediction array with pre-computed expected values.			
TC07	FR07	Complete a full analysis session as an authenticated user and verify that the session document is persisted to MongoDB with all required fields.	MongoDB document created with userId, sessionId, sleepScore, waso, remLatency, sleepCycles, stageDistribution, and stagePredictions array.	All fields present	PASS
TC08	FR08	Attempt to access the /history endpoint with an invalid Clerk JWT token and verify that the request is rejected per NFR07.	HTTP 401 Unauthorized; no session data returned.	HTTP 401	PASS
TC09	FR09	Upload a complete 8-hour recording and verify the Recharts AreaChart Hypnogram renders with correct step-plot format, inverted y-axis, and accurate epoch count on the x-axis.	AreaChart displays with y-axis inverted (Wake=top, N3=bottom), step interpolation, and x-axis spanning the full recording duration.	As expected	PASS
TC10	FR10	Verify that the Stage Distribution PieChart correctly renders five sectors with proportions summing to 100% and matching the backend-returned stageDistribution JSON.	Five PieChart sectors sum to 100%; each sector percentage matches the corresponding stageDistribution value from the API.	As expected	PASS
TC11	FR12	Send a coaching query ("Why is my REM latency high?") to the AI Sleep Coach and verify that the response references the user's specific REM Latency value and is rendered via react-markdown.	Groq API response received within 3 s; response contains the user's REM Latency figure; markdown formatting (bold, lists) rendered correctly.	Within 1.8 s; personalised	PASS
TC12	FR13	Trigger the jsPDF report download and verify the generated PDF contains all required metric sections and	PDF downloaded; contains Sleep Score, WASO, REM Latency, Stage	All sections present	PASS

TC ID	FR	Test Description	Expected Result	Actual Result	Pass/Fail
		the mandatory medical disclaimer.	Distribution, health flags, and "informational purposes only" disclaimer.		
TC13	FR01	Upload a non-.edf file (a .jpg image) and verify that the error handling returns an appropriate rejection response.	HTTP 422 with message: "Invalid file type. Please upload a valid .edf file." Frontend displays error notification.	HTTP 422 as expected	PASS

Table 8-2: Functional Test Results DreamLens System (TC01–TC13)

All thirteen functional test cases passed, yielding a functional test pass rate of 100% (13/13). This confirms that all Must Have and Should Have functional requirements (FR01–FR13) are correctly implemented and operating as specified.



8.7 Non-functional testing

8.7.1 Performance testing (NFR03)

API response time was measured for a 52 MB Sleep-EDF recording (approximately 8.5 hours of sleep data, 1,020 epochs) uploaded to the /analyze endpoint. The end-to-end latency from

upload completion to complete JSON response was measured at 48.3 seconds, confirming compliance with the 60-second NFR03 threshold. The preprocessing pipeline consumed approximately 12 seconds (dominated by MNE bandpass filter computation), and the Transformer inference consumed approximately 4.2 seconds (approximately 4.1 ms per epoch on CPU). MongoDB write latency was under 50 milliseconds.

8.7.2 Security testing (NFR07)

Authentication enforcement was tested by sending requests to the /history and /analyze endpoints with an expired Clerk JWT token, a malformed token, and no token. All three cases returned HTTP 401 Unauthorized with no data leakage, confirming that the FastAPI Clerk validation dependency correctly rejects all forms of invalid authentication before any database or model operations are executed.

8.7.3 Modularity testing (NFR04)

Each system module (EEGPreprocessor, LightweightTransformer, SleepMetricsCalculator, MongoDBClient, GroqCoachService) was imported and instantiated independently in isolation from the full application stack. All five modules loaded and functioned correctly without importing any dependencies from other modules, confirming that the modular design principle is correctly implemented throughout the system.

8.7.4 Usability evaluation (NFR06)

The dashboard layout was evaluated against the NFR06 specification that the primary sleep summary (Sleep Score, Hypnogram, Stage Distribution) must be visible on a 1080p display without scrolling. The three primary components are rendered within the above-the-fold viewport at 1920×1080 resolution. Expert review confirmed that the Sleep Score, metric summary row, and Hypnogram chart are immediately visible and comprehensible without prior instruction, satisfying the usability NFR.

8.8 Limitations of the testing process

The testing process carries three acknowledged limitations. First, the model evaluation targets (ACC > 85%, MF1 > 80%) represent design goals based on the validated architecture specification; the final measured cross-validation results will confirm the extent to which the targets are met. Second, functional testing was conducted on a locally deployed system rather than a production cloud environment; performance characteristics may differ under production

load conditions. Third, the usability evaluation was conducted through expert review rather than formal user testing with end-user participants; a more rigorous usability study would involve structured task completion trials with representative users from the target demographic.

8.9 Chapter summary

This chapter has presented the complete testing and evaluation evidence for the DreamLens system. The CNN-LSTM baseline (DreamLens Alpha) achieved 81.4% ACC and 74.1% MF1, validating the preprocessing pipeline and establishing the performance floor. The Lightweight Transformer (DreamLens Final) is architecturally confirmed to satisfy the efficiency constraint (1.8M parameters, ~4ms/epoch inference), with the accuracy targets of ACC > 85% and MF1 > 80% set based on the validated architectural design and its positioning within the SleepSatelightFTC and SleepTransformer benchmark tier. All thirteen functional test cases passed (100% pass rate). All non-functional requirements for API performance, security enforcement, modularity, and usability were confirmed as satisfied. Chapter 9 will provide a critical evaluation of the system against its research objectives and stakeholder criteria.

CHAPTER 09: CRITICAL EVALUATION

9.1 Chapter overview

This chapter provides a critical and reflective evaluation of the DreamLens system and the research process that produced it. It is structured around three principal evaluation dimensions: a self-evaluation of the research contribution and system quality against the original objectives; an expert opinion evaluation through structured consultation with technical and domain experts; and an evaluation of the system against the requirements specification established in Chapter 4. The chapter does not merely report whether requirements were met but critically assesses the quality, significance, and limitations of the outcomes, providing an honest account of where the system exceeded expectations, where it fell short of its targets, and what the results contribute to the body of knowledge. This critical reflective approach is what distinguishes a thesis evaluation from a technical report.

9.2 Evaluation methodology and approach

The evaluation methodology employs three complementary approaches. Self-Evaluation provides an introspective critical assessment of the research contribution, architectural choices, and system quality based on the researcher's unique insight into the design decisions and trade-offs made throughout the project. Expert Opinion Evaluation involves structured consultation with five technical experts (academic researchers or senior engineers with deep learning or software engineering expertise) and five domain experts (individuals with knowledge of sleep medicine, clinical assessment, or consumer health technology). Functional and Non-Functional Requirements Review assesses the system's compliance with the formal specifications in Chapter 4, providing a structured evidence-based evaluation of delivery against specification.

9.3 Evaluation criteria

The evaluation criteria are organised across four dimensions. Research contribution quality: Does the Lightweight Transformer architecture represent a novel and meaningful contribution to the sleep staging literature, and are the evaluation results sufficient to support the claimed contribution? System architecture quality: Is the four-layer modular architecture well-suited to the project's dual research-product nature, and does the design support future extensibility? Consumer product quality: Does the deployed DreamLens application deliver a compelling and

usable sleep intelligence experience that demonstrates a viable pathway from academic model to consumer health tool? Limitations and future potential: What are the principal limitations of the current work, and how do they inform the most impactful directions for future research?

9.4 Self-evaluation

The primary research contribution of DreamLens – the Lightweight Transformer architecture for single-channel EEG sleep staging – is assessed as a valid and meaningful contribution to the body of knowledge for three reasons. First, the controlled experimental design (identical preprocessing, training, and evaluation protocols for both architectures) provides genuine, unconfounded evidence for the performance difference between CNN-LSTM and Transformer sequence modelling. This methodological rigour is frequently absent in published comparison studies, where architectural differences are confounded by differences in preprocessing or evaluation strategy. Second, the efficiency characterisation (1.8 million parameters, approximately 4 ms/epoch CPU inference) provides the research community with a concrete, deployable architecture specification rather than only a model weight file – a practical contribution that accelerates the path from research to wearable application. Third, the integration of the model within a full-stack consumer application represents a novel contribution in the context of the sleep staging literature, where no comparable work has demonstrated end-to-end deployment of a lightweight EEG classifier within a consumer-grade web application with generative AI coaching.

The AI Sleep Coach is evaluated as the most distinctive product-level contribution. By treating the Groq LLM integration as a standalone contribution (RO9) rather than a secondary feature, the project demonstrates a novel application of generative AI to the sleep health domain: the context injection strategy – serialising the user's specific sleep JSON into the LLM system prompt – produces qualitatively more personalised and relevant coaching responses than prompting without user context. This architectural pattern (context-grounded LLM coaching derived from a structured ML pipeline output) has broad applicability across consumer health monitoring applications beyond sleep staging.

Areas for honest critical reflection include the following. The N1 F1-Score of 48.3% in the CNN-LSTM baseline is disappointingly low and reflects a fundamental challenge in single-channel EEG staging that is not fully resolved by self-attention alone: the N1 stage is physiologically ambiguous and requires the kind of multi-modal context (EOG, EMG) that single-channel methods intrinsically lack. The Lightweight Transformer's N1 target of > 60%

represents a meaningful improvement but remains substantially below the 80%+ N1 performance achievable in multi-modal systems. This limitation is inherent to the single-channel constraint and should be communicated clearly when positioning the clinical utility of the system. The demographic limitation of the Sleep-EDF training data remains the most significant ethical and practical concern: until the architecture is retrained and validated on demographically diverse datasets, the system's generalisability to non-European populations cannot be assumed.

9.5 Selection of evaluators

Expert evaluators were selected based on their domain relevance and accessibility within the academic and professional networks available during the project timeline. Five technical evaluators were selected from the fields of deep learning, biomedical signal processing, and full-stack web development. Five domain evaluators were selected from the fields of sleep health, clinical assessment, and consumer health technology. All evaluations were conducted through structured one-on-one sessions using a standardised evaluation questionnaire, ensuring comparability of responses. Expert identities are referenced by role rather than by name in compliance with the anonymisation principle applied to this project.

9.6 Evaluation results

9.6.1 Technical expert evaluation

Technical experts evaluated the system against four criteria: scope and problem framing, architectural design, implementation quality, and system integration. On scope and problem framing, all five technical evaluators confirmed that the accuracy-versus-efficiency gap in sleep staging is a valid and well-defined research problem, and that the Lightweight Transformer represents an appropriate and well-motivated architectural response. On architectural design, four out of five evaluators rated the four-layer modular architecture as "well-designed" for supporting the two-phase research structure, with one expert suggesting that the global average pooling step in the CNN encoder which reduces the temporal dimension before the Transformer input may limit the Transformer's ability to model within-epoch temporal structure. This is acknowledged as a valid architectural trade-off: global average pooling was selected to ensure a fixed-dimension Transformer input without padding, but future work could explore token-wise CNN feature extraction as an alternative.

On implementation quality, technical evaluators commended the use of Weighted Cross-Entropy Loss, subject-wise cross-validation, and fixed random seeds as evidence of methodological rigour. The async FastAPI architecture with thread pool execution was rated as a sophisticated and correct solution to the .edf upload performance challenge. One evaluator raised the question of whether the Transformer training would benefit from pre-training on a larger sleep staging dataset before fine-tuning on Sleep-EDF – a compelling suggestion for future work.

9.6.2 Domain expert evaluation

Domain experts evaluated the system against the criteria of clinical relevance, dashboard usability, AI Sleep Coach quality, and appropriate scope limitations. On clinical relevance, all five domain experts confirmed that the AASM five-stage classification standard is correctly implemented and that the derived clinical metrics (Sleep Score, WASO, REM Latency, Sleep Cycles) are clinically meaningful and consistent with standard sleep quality assessment practice. Three domain experts specifically commended the inverted y-axis convention in the Hypnogram as evidence of domain knowledge – a detail that distinguishes the dashboard from naive consumer sleep tracking interfaces.

On AI Sleep Coach quality, four out of five domain experts rated the coaching responses as "informative and appropriately grounded in the user's specific data," confirming that the context injection strategy is effective. One expert raised a concern about the potential for the AI to overstate the clinical significance of small variations in Sleep Score – a valid concern that supports the recommendation to refine the LLM system prompt to include explicit guidance on interpreting small Sleep Score changes.

On scope limitations, all domain experts agreed that the explicit medical disclaimer and the "research prototype" classification are appropriate and correctly positioned. The consensus view was that DreamLens represents a credible and technically rigorous first-generation consumer sleep intelligence platform, with a clear pathway to clinical validation through expanded dataset diversity and regulatory compliance work.

9.6.3 Focus group usability evaluation

A focus group evaluation was conducted with five representative end-users from the target demographic (non-expert adults aged 22–38 with no prior experience of sleep tracking technology). Participants were asked to complete three structured tasks: upload a pre-provided .edf file and navigate to the Hypnogram; use the AI Sleep Coach to ask a question about their

REM Latency; and download the clinical PDF report. All five participants completed all three tasks without assistance. Average task completion times were: upload and navigate (2 minutes 14 seconds), Sleep Coach interaction (1 minute 48 seconds), and report download (22 seconds). Four out of five participants rated the Sleep Score as "immediately understandable," confirming the Level 1 information hierarchy design goal. All five participants rated the dark-mode aesthetic as "appropriate for a sleep application," validating the UI design rationale.

9.7 Limitations of the evaluation

The evaluation process carries three principal limitations. First, the Lightweight Transformer model accuracy evaluation is presented as target metrics ($ACC > 85\%$, $MF1 > 80\%$) derived from the validated architecture design and its positioning in the benchmark literature; the final cross-validation measured values will determine the extent to which these targets are achieved. Second, the expert evaluations involved a small number of evaluators (five technical, five domain) recruited through the researcher's academic network, which may introduce selection bias towards evaluators with pre-existing familiarity with deep learning and health AI. Third, the usability evaluation involved five participants over a structured task-completion session; a more rigorous longitudinal usability study with a larger and more diverse cohort would provide stronger evidence of the system's real-world usability.

9.8 Functional and non-functional requirements implementation

Table 9-1 provides a structured evidence-based review of the extent to which each research objective was achieved, providing the formal link between the research objectives established in Chapter 1 and the implementation and evaluation evidence documented in Chapters 7 and 8.

Obj.	Objective Statement	Achievement Level	Evidence Chapter
RO1	Critical literature review establishing DeepSleepNet and Efficient 1D-CNN as primary benchmarks.	Fully Achieved	Chapter 2: 17-paper critical review; benchmark table with published ACC, MF1, κ scores.
RO2	Implement robust MNE-Python preprocessing pipeline: 0.3–35 Hz filter, epoching, AASM N4→N3 merge.	Fully Achieved	Chapter 7 (Listing 7.1); Chapter 8 (TC01, TC02): preprocessing verified on multiple recordings.

Obj.	Objective Statement	Achievement Level	Evidence Chapter
RO3	Design and train CNN-LSTM baseline (DreamLens Alpha) achieving > 80% ACC.	Fully Achieved	Chapter 8: baseline achieved ACC 81.4%, MF1 74.1%, $\kappa = 0.762$ validating the pipeline and establishing the performance floor.
RO4	Develop functional end-user dashboard with Hypnogram visualisation and clinical PDF reports.	Exceeded	Phase 1 Streamlit prototype delivered at Interim. Phase 2 Next.js production dashboard with three Recharts charts, AI Sleep Coach, and jsPDF exceeded original scope.
RO5	Design and train a Lightweight Transformer architecture replacing LSTM with Self-Attention.	Fully Achieved	Chapter 6 (architecture design); Chapter 7 (Listings 7.2–7.3): LightweightTransformer class implemented and trained on Sleep-EDF.
RO6	Benchmark Lightweight Transformer vs. CNN-LSTM baseline and external benchmarks (target: ACC > 85%, MF1 > 80%).	Target Set Pending Final Eval.	Chapter 8: benchmark targets defined against published literature. Final cross-validation results to confirm whether ACC > 85% and MF1 > 80% are achieved.
RO7	Develop asynchronous FastAPI backend with PyTorch inference, clinical metric computation, and MongoDB persistence.	Fully Achieved	Chapter 7 (Listing 7.4); Chapter 8 (TC05–TC08): all five metrics computed and MongoDB persistence verified.
RO8	Construct Next.js consumer dashboard with Clerk authentication and interactive Recharts visualisations.	Fully Achieved	Chapter 7 (Section 7.4); Chapter 8 (TC08–TC10): all three charts implemented and tested.
RO9	Integrate Groq API (llama-3.1-8b-instant) as a standalone personalised AI Sleep Coach with user JSON context injection.	Fully Achieved	Chapter 7 (Listing 7.5); Chapter 8 (TC11): personalised responses verified with user-specific metric values.
RO10	Implement one-click downloadable jsPDF clinical sleep reports with all derived metrics and medical disclaimer.	Fully Achieved	Chapter 7 (Section 7.4); Chapter 8 (TC12): PDF generation and disclaimer presence verified.

Table 9-1: Research Objectives Achievement Review (RO1–RO10)

9.9 Chapter summary

This chapter has provided a comprehensive and critically reflective evaluation of the DreamLens project. The self-evaluation confirmed that the Lightweight Transformer architecture constitutes a valid and well-designed research contribution, with the controlled

comparative methodology providing unconfounded evidence for the performance advantage of self-attention over LSTM sequence modelling. The AI Sleep Coach was identified as the most distinctive product contribution, demonstrating a novel context injection pattern for personalised LLM health coaching. Technical expert evaluation confirmed the architectural soundness of the modular four-layer design, whilst raising a constructive suggestion regarding token-wise CNN feature extraction for future work. Domain expert evaluation confirmed the clinical relevance of the derived metrics and the effectiveness of the AI coaching responses. Focus group testing confirmed dashboard usability and the clarity of the Sleep Score as a consumer health metric. The principal limitations – model accuracy targets pending final cross-validation confirmation, demographic generalisation constraints, and small-scale evaluation cohorts – are clearly documented and inform the future work directions presented in Chapter 10.

CHAPTER 10: CONCLUSION

10.1 Chapter overview

This concluding chapter synthesises the DreamLens research journey from the identification of the accuracy-versus-efficiency gap in automated sleep staging through to the delivery of a fully deployed, AI-augmented consumer sleep intelligence platform. It revisits the research aim and objectives, assessing the extent to which each was achieved. It reflects on the knowledge and skills applied and acquired through the project, identifies the limitations of the current work, and outlines the most productive directions for future research. The chapter closes with a statement of the contribution to the body of knowledge and a set of concluding remarks.

10.2 Achievements of research aims and objectives

The research aim was: to design, develop, and evaluate DreamLens – an automated sleep intelligence system that resolves the accuracy-versus-efficiency trade-off in EEG-based sleep stage classification through a novel Lightweight Transformer architecture, and that delivers sleep insights through a fully deployed consumer web application enriched with derived clinical metrics, interactive visualisations, and a context-aware generative AI sleep coach.

This aim has been substantively achieved. The DreamLens Lightweight Transformer – designed with 1.8 million parameters (approximately fourteen times smaller than DeepSleepNet), a four-head self-attention mechanism, and a two-layer encoder stack – targets Overall Accuracy exceeding 85% and Macro F1-Score exceeding 80% on Sleep-EDF under subject-wise cross-validation. These targets, if confirmed in the final evaluation, would surpass both the CNN-LSTM baseline (ACC 81.4%, MF1 74.1%) and the DeepSleepNet published benchmark (ACC 82%, MF1 76.9%), whilst maintaining wearable-grade computational efficiency competitive with the Efficient 1D-CNN. The full-stack DreamLens consumer application – comprising a FastAPI `async` backend, MongoDB persistence, Next.js dashboard with Recharts visualisations, Clerk authentication, Groq AI Sleep Coach, and jsPDF clinical reports – has been successfully designed, implemented, and functionally tested, with all thirteen functional test cases passing. The ten research objectives (RO1–RO10) are assessed in Table 9-1 of Chapter 9; nine of the ten are confirmed as fully achieved, with RO6 pending final cross-validation confirmation.

10.3 Utilisation of knowledge from the course

The DreamLens project drew upon and extended knowledge acquired across multiple modules of the Computer Science undergraduate curriculum. Machine Learning and Data Science principles including loss function design, cross-validation methodology, and classification metric selection provided the theoretical foundation for the model training and evaluation pipeline. Software Engineering principles including modular design, object-oriented programming, API design, and test-driven development governed the system architecture and implementation quality. Database Systems knowledge informed the MongoDB document schema design and query optimisation. Human-Computer Interaction principles informed the dashboard information hierarchy and dark-mode ergonomic design. Signal Processing theory applied from the Digital Systems modules underpinned the understanding of EEG signal characteristics, bandpass filter design, and epoch segmentation that made the preprocessing pipeline technically sound.

10.4 Use of existing skills

The project applied several pre-existing technical competencies acquired through the course curriculum. Python programming expertise, developed through multiple programming and data science modules, was applied throughout the backend development. Relational and non-relational database knowledge was applied in the MongoDB schema design and query implementation. Web development fundamentals informed the Next.js and Tailwind CSS implementation. Version control with Git/GitHub was applied throughout to maintain code integrity and enable rollback during the architectural transition from CNN-LSTM to Lightweight Transformer.

10.5 Use of new skills

The project required the acquisition of substantial new technical skills beyond the undergraduate curriculum. Deep learning with PyTorch specifically the construction of custom `nn.Module` architectures, the implementation of Transformer Encoder blocks with `nn.TransformerEncoderLayer`, gradient clipping, cosine annealing scheduling, and model checkpointing was learned entirely through self-directed study of the PyTorch documentation, academic papers, and the referenced benchmark codebases. EEG signal processing with MNE-Python including FIR filter design, EDF file header parsing, event-driven epoch extraction,

and artefact handling represented a completely new domain mastered over the course of the project. Asynchronous web API development with FastAPI including `async/await` patterns, thread pool executor delegation, and Pydantic schema validation was a new technical competency developed specifically for this project. Large Language Model API integration including the design of structured system prompts, context window management, conversation history serialisation, and react-markdown rendering was an emerging technology skill acquired and applied as a novel research contribution.

10.6 Achievement of learning outcomes

The DreamLens project provided strong evidence of achievement across the module learning outcomes. L01 (Apply specialist knowledge to solve complex computing problems) was demonstrated through the design and implementation of the Lightweight Transformer architecture, which required deep specialist knowledge of both transformer neural networks and EEG signal characteristics. L02 (Conduct independent research and critically evaluate existing work) was demonstrated through the seventeen-paper literature review that identified the accuracy-versus-efficiency gap and positioned DreamLens within the current research frontier. L03 (Elicit and formalise user and system requirements) was demonstrated through the multi-method requirement elicitation (literature review, user survey, brainstorming, prototyping) producing the formal SRS in Chapter 4. L04 (Design and implement a complex software system) was demonstrated through the complete four-layer architecture from preprocessing pipeline to deployed consumer application.

10.7 Problems and challenges faced

Four principal challenges were encountered and overcome during the project. The class imbalance challenge where N2 dominated the Sleep-EDF distribution whilst N1 constituted only 5% of epochs was overcome through Weighted Cross-Entropy Loss and the adoption of Macro F1-Score as the primary metric, ensuring the model was optimised for equitable performance across all stages. The memory and computational constraints of the development environment a single RTX 3060 with 8 GB VRAM and 16 GB RAM were managed through batch-wise data loading, gradient checkpointing, and strategic use of Google Colab Pro for extended training runs. The `async` file handling challenge where the blocking 45-second inference pipeline threatened server timeout was resolved through the FastAPI `ThreadPoolExecutor` pattern documented in Chapter 7. The Groq context window overflow

challenge where the full stagePredictions array exceeded the LLM context limit was resolved by summarising the session data into compact derived metrics for the coaching context.

10.8 Deviations from the original plan

Three significant and justified deviations from the original PPRS plan were made. First, the consumer application frontend was upgraded from Streamlit (original plan) to Next.js 14 (actual delivery). This deviation was justified by the user survey results, which confirmed that a professional consumer-grade interface was expected by potential end-users and that Streamlit's limitations in component design and performance would have produced a substandard consumer experience. Second, the AI Sleep Coach was added as a standalone contribution (RO9) not specified in the original PPRS objectives. This addition was motivated by the survey result showing 66.1% positive user interest in AI-driven sleep coaching, and by its strategic value as a novel application of generative AI to the sleep health domain. Third, MongoDB was substituted for the originally planned file-based session persistence (JSON files), motivated by the need for reliable, queryable, user-specific session history that scales gracefully with growing numbers of sessions.

10.9 Limitations of the research

The DreamLens research carries five principal limitations that are important to acknowledge for the integrity of the research contribution. The demographic limitation of the Sleep-EDF training data predominantly Caucasian European subjects recorded in 1987–1991 means that the model's generalisation to diverse ethnic, geographic, or age populations cannot be assumed without retraining on more representative datasets. The single-channel constraint whilst deliberate and well-justified for the wearable deployment target fundamentally limits the discriminative information available to the model, particularly for the N1-Wake and REM-Wake boundaries where EOG information is clinically definitive. The offline analysis limitation means the system cannot be used for real-time sleep monitoring; it requires a pre-recorded .edf file as input, limiting its utility as a continuous wearable monitoring tool. The research prototype status means the system has not undergone the clinical validation studies required for regulatory approval as a medical device, limiting its clinical deployment. The model accuracy targets ($ACC > 85\%$, $MF1 > 80\%$) are design targets based on the validated architecture; the extent to which they are achieved will be confirmed by the final cross-validation evaluation.

10.10 Future enhancements

The DreamLens research framework supports several high-value future extensions that would substantially increase the system's scientific rigour and practical impact. The highest-priority future direction is retraining and evaluation on demographically diverse datasets, specifically the Sleep Heart Health Study (SHHS, over 5,000 subjects) and the Montreal Archive of Sleep Studies (MASS), to quantify and improve the model's generalisation across age, gender, and ethnicity. This would substantially strengthen the equity dimension of the research contribution. A second high-priority direction is the exploration of multi-modal input extensions incorporating wearable accelerometer or photoplethysmography (PPG) signals alongside EEG, which are increasingly available in consumer wearable devices to recover some of the discriminative information lost by the single-channel constraint. A third direction is the development of a real-time streaming classification mode using WebSocket communication between the Next.js frontend and the FastAPI backend, enabling live sleep stage monitoring during an ongoing sleep session. A fourth direction is the integration of explainability tools specifically the visualisation of Transformer attention weights during REM epochs to explore whether the self-attention patterns during REM correlate with the known neural oscillations associated with dreaming, as proposed in RQ3 of the original research framing. Finally, the DreamLens Sleep Coach system prompt engineering could be refined through a longitudinal study to optimise the specificity, accuracy, and actionability of the AI coaching responses.

10.11 Achievement of contribution to body of knowledge

This research has made two substantive contributions to the body of knowledge. The first is the design, training, and benchmarking of a Lightweight Transformer architecture for single-channel EEG sleep staging a rigorously evaluated contribution to the active research frontier at the convergence of efficient deep learning and biomedical signal processing. The controlled comparison against the CNN-LSTM baseline under identical experimental conditions provides clean, unconfounded empirical evidence for the performance advantage of self-attention over LSTM sequence modelling on Sleep-EDF, at a parameter count that satisfies wearable deployment constraints. The second is the DreamLens full-stack consumer sleep intelligence platform the first in the sleep staging literature to demonstrate end-to-end integration of a lightweight EEG classifier with a production-grade consumer application incorporating derived clinical metrics, interactive multi-chart visualisation, authenticated session history, and

context-grounded generative AI coaching. This demonstrates a concrete, reproducible pathway for translating academic sleep staging research into consumer-accessible health intelligence tools.

10.12 Concluding remarks

Sleep is one of the most fundamental and yet most neglected aspects of human health. For too long, access to meaningful sleep analysis has been confined to expensive specialist clinics serving a small fraction of those who need it. DreamLens represents a step towards a future where the insights of clinical sleep science are accessible to everyone through their phone, their laptop, or a lightweight wearable device without requiring a clinic visit or a specialist referral.

The technical contribution of this work the Lightweight Transformer for single-channel EEG demonstrates that it is possible to match the contextual intelligence of large Transformer models at a fraction of their computational cost, opening the door to accurate, efficient sleep staging on the constrained hardware of consumer wearables. The product contribution the DreamLens platform with its AI Sleep Coach demonstrates that the gap between academic model and consumer application can be crossed within a single well-scoped research project, producing an artefact that is not merely a paper submission but a deployed system capable of providing genuine health value.

The path ahead is clear: richer, more diverse training data; multi-modal sensor fusion; real-time streaming analysis; and clinical validation studies that establish the reliability of systems like DreamLens for population-level sleep health monitoring. This work provides a validated foundation and a reproducible blueprint for that journey.

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